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# A unified model for species fluxes, ion impact energy and heat transfer at the electrode of an atmospheric-pressure arc discharge

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## Abstract

## 1 Introduction

Arc discharges find extensive applications in welding, cutting, and thermal spraying. In recent years, their use has expanded to emerging applications, including plasma pyrolysis, advanced material synthesis, and metallurgical reduction processes [1, 2, 3]. An atmospheric-pressure arc discharge spans a wide range of length scales, differing by several orders of magnitude. The millimeter-to-centimeter-scale quasi-neutral arc column constitutes the bulk of the discharge, where convection, thermal conduction, Joule heating, and radiation interact under conditions that are often close to local thermodynamic equilibrium (LTE). In contrast, the sub-micron to sub-millimeter regions adjacent to the electrodes are characterized by strong nonequilibrium effects. In these regions, the plasma is subjected to the relatively cold electrode surfaces, steep density gradients associated with particle absorption at the electrodes, and strong electric fields arising from charge separation. These fields can accelerate electrons and ions to energies substantially higher than their local thermal energies.

Resolving the full range of length scales throughout the entire discharge is neither computationally practical nor necessary for many applications. Consequently, arc-discharge models commonly treat the quasi-neutral bulk plasma separately from the nonequilibrium electrode regions. In the bulk, the quasineutrality condition, $n_i \approx n_e$, eliminates the need to explicitly solve Poisson's equation, while the near-LTE assumption further reduces model complexity by allowing thermodynamic and transport properties to be determined primarily from the local temperature and pressure. Macroscopic effects such as cathode jets, electrode geometry, convection, and electromagnetic forces can then be incorporated within a continuum formulation based on the Navier–Stokes equations with Joule-heating and Lorentz-force terms. The non-equilibrium electrode regions are generally modeled to account for diffusion, reactions and sheath effects.

Quasi-neutral bulk plasma models can be broadly classified into four categories according to the degree of thermodynamic and chemical equilibrium assumed and the treatment of electron and heavy-particle temperatures.

*(i) Full local thermodynamic equilibrium.* Experimental measurements of species temperatures and electron densities indicate that the arc core is generally close to LTE [4, 5, 6]. In a full LTE model, electrons and heavy particles are characterized by a single temperature $T$, with Maxwellian velocity distributions, Boltzmann excitation, Saha ionization, and Dalton's law used to determine the plasma state and properties. The governing equations typically consist of mass, momentum, energy, and current continuity [7, 8, 9]. Conventional LTE models do not account for electrode falls and sheath effects.

*(ii) Two-temperature models.* Two-temperature (2T) models treat electrons and heavy particles as separate Maxwellian populations at $T_e$ and $T_h$, coupled through elastic energy exchange. In two-temperature-equilibrium (2T-EQ) models, the chemical composition is additionally determined from a generalized Saha relation for $T_e \neq T_h$. Hsu and Pfender [10] showed that $T_e \approx T_h$ in the high-density arc core, with significant temperature separation emerging in the low-density fringes and near the anode. However, the generalized Saha closure is not unique: different formulations can predict electron densities differing by up to an order of magnitude away from $T_e/T_h \approx 1$ [11].

Furthermore, inconsistencies in the partitioning of ionization-energy and reactive thermal-conductivity terms between the electron and heavy-particle energy equations have been identified by Freton et al. [12] in several formulations.

*(iii) Nonequilibrium models.* Fully nonequilibrium models abandon equilibrium closures such as LTE and generalized Saha equilibrium, allowing the plasma composition and electron and heavy-particle temperatures to evolve independently. These models typically include multicomponent species transport and separate electron and heavy-particle energy equations. Examples include the nonequilibrium arc models of Haidar [13], Gonzalez *et al.* [14], and Amakawa *et al.* [15]. By resolving departures from equilibrium directly, these models can describe the transition from the LTE-like bulk plasma toward the strongly nonequilibrium electrode region. However, the space-charge sheath is generally still treated separately using an appropriate sheath model.

*(iv) LTE with diffusion-driven ionization nonequilibrium.* Another approach considered in the literature retains $T_e = T_h = T$ while allowing the electron and ion densities to depart from their local Saha-equilibrium values [16, 17, 18]. This departure arises when diffusion competes with the ionization–recombination relaxation processes, particularly near the electrodes. Lowke [16] demonstrated this approach using a full-diffusion calculation that solves the ambipolar electron-continuity equation together with current continuity, including the electron-diffusion contribution. The resulting solution produces a retarding-field potential structure near the electrode that is absent in a conventional LTE calculation. However, because $T_e = T_h$ is retained throughout the domain, this formulation cannot capture the strong electron–heavy-particle temperature separation expected in the low-density electrode-adjacent region.

### *1.1 Coupling the bulk plasma to the electrode: sheath models*

Regardless of the bulk-plasma model employed, the bulk solution must ultimately be coupled to a description of the electrode region. The appropriate coupling strategy depends on the assumptions underlying the bulk formulation and, in particular, on the degree of nonequilibrium that is retained. For fully LTE models (i), the electrode region must account for the breakdown of LTE, including the separation between electron and heavy-particle temperatures, deviations from Saha equilibrium, and the formation of a space-charge sheath. For two-temperature and two-temperature-equilibrium models (ii), the bulk formulation already accounts for electron–heavy-particle temperature nonequilibrium; however, deviations from ionization equilibrium and the formation of a space-charge sheath must still be described. In fully nonequilibrium bulk models (iii), an additional ionization layer or presheath is not necessarily required, since departures from equilibrium are already permitted within the bulk formulation. The principal remaining transition to the electrode is therefore associated with the breakdown of quasineutrality and the formation of the space-charge sheath.

A common approach is to construct the electrode-region model from separate submodels for the ionization layer or presheath and the sheath. The ionization layer is treated as a collisional region and coupled to the sheath, which is treated as collisionless using an analytical model [19, 20, 21, 22]. This approach avoids the direct solution of the stiff Poisson equation across the sheath. However, Benilov *et al.* [23] pointed out that such matching is strictly valid only if the sheath is collision-dominated.

Almeida, Benilov, and Naidis [24] evaluated the number of collisions an ion undergoes as it traverses the cathode space-charge sheath, and showed that in an atmospheric-pressure argon arc this number is far greater than unity rather than $\ll 1$. A physically consistent electrode model for high-pressure arc discharges should therefore account for collisional ion transport within the sheath. A kinetic study of the near-cathode region by Sun et al. [25] supported this collision-dominated view. Further, Semenov et al. [26] demonstrated that simple sheath models that assume the Bohm velocity at the sheath edge are not valid for the anode region of high-pressure arc discharges.

An alternative approach is *unified modeling*, in which a single set of governing equations, including Poisson's equation, is solved across the entire interelectrode region without explicitly separating the bulk plasma from the sheath. This approach provides a self-consistent description of the electrode region but is computationally expensive because the mesh must resolve the extremely small sheath length scales. Consequently, applications to arc discharges have largely been restricted to one-dimensional calculations extending from the electrode into the bulk plasma [24, 27, 26].

Except in the immediate vicinity of the electrodes, an atmospheric-pressure arc is sufficiently close to local thermodynamic equilibrium (LTE) that the bulk plasma can be modeled using a single temperature. Consequently, nonequilibrium effects need not be considered in the bulk, which substantially reduces the complexity of the model. The approach adopted in the present work

therefore retains the computationally efficient LTE treatment for the bulk of the arc while explicitly resolving the nonequilibrium physics in the near-electrode region. Rather than prescribing electrode-boundary quantities or relying on an analytical sheath model, a unified drift-diffusion–Poisson formulation is employed to simultaneously determine the charged-particle distributions, species fluxes, ion impact energies, heat transfer to the electrode, and sheath voltage drop. The same governing drift-diffusion–Poisson system is solved continuously from the quasi-neutral bulk plasma through the near-electrode transition region and into the sheath. The highly disparate spatial scales are resolved using adaptive mesh refinement (AMR), with the computational mesh refined to a fraction of the local Debye length in the sheath region.

An additional advantage is that field reversals need not be anticipated. Field reversals and changes in the sign of the potential drop near the anode have been reported in atmospheric-pressure arcs and can depend on several factors, including the attachment mode, current density, and electrode ablation effects [28, 29, 30]. By solving the drift-diffusion–Poisson system self-consistently across the near-electrode region, the sheath potential structure and its polarity emerge from the coupled solution rather than being assumed a priori. This approach enables one to use the results from the computationally efficient bulk plasma treatment to inform the boundary conditions for the near-electrode region, while simultaneously capturing the nonequilibrium effects that arise in the sheath and ionization layer. The model and its numerical implementation in the PlasmeX code are described in Section 2.

## 2 Physical model and scale analysis

### *2.1 Governing equations*

In the electrode region, the plasma is in contact with a relatively cold, absorbing electrode surface. This interaction produces strong gradients in the plasma properties, and the plasma departs progressively from LTE as the electrode is approached. We estimate the length scales over which these departures develop by considering the LTE plasma adjacent to the electrode region at temperatures between $T = 12{,}000$ K and $T = 22{,}000$ K, which are typical of the anode and cathode regions, respectively, of a high-intensity atmospheric-pressure arc discharge [16, 9, 31].

In the bulk plasma, the electron density $n_e$ is governed by ionization and recombination reactions. The characteristic time scale of these reactions can be obtained by linearizing $\dot{n}_e = k_{\mathrm{ion}} n_e n_n - k_{\mathrm{rec}} n_e^3$ (with $n_i = n_e$) about equilibrium at fixed $p$ and $T$ (so that $\delta n_n = -2\,\delta n_e$), which gives $\tau_{rxn} = 1/(2k_{\mathrm{rec}} n_e^2 + 2k_{\mathrm{ion}} n_e)$. A characteristic length scale for the reactions is then $\lambda_{rxn} = \sqrt{D_a \tau_{rxn}}$, where $D_a$ is the ambipolar diffusion coefficient. This length scale implies that a plasma out of chemical equilibrium needs a distance of order $\lambda_{rxn}$ to relax back to equilibrium. Similarly, since an electron loses only a fraction $2m_e/M$ of its energy in an elastic collision with a heavy particle, the length scale for the energy relaxation between electrons and heavy species can be estimated as $\lambda_T = 1/[\sqrt{2m_e/M}\,(n_i \overline{Q}_{ei} + n_n \overline{Q}_{en})]$, where $n_i$ and $n_n$ are the ion and neutral number densities, and $\overline{Q}_{ei}$ and $\overline{Q}_{en}$ are the averaged momentum-transfer cross sections for electron–ion and electron–neutral collisions. $\overline{Q}_{en}$ is the Maxwellian average of the argon momentum-transfer cross section used in Appendix A. Electron–ion collisions are Coulomb collisions, for which the averaged momentum-transfer cross section is $\overline{Q}_{ei} = e^4 \ln\Lambda/[32\pi\varepsilon_0^2 (k_B T_e)^2]$ [24]. Here, $k_B$ is the Boltzmann constant and $\ln\Lambda$ is the Coulomb logarithm, $\ln\Lambda = \ln\left(1.24 \times 10^7 T_e^{3/2} n_e^{-1/2}\right)$, with $T_e$ in K and $n_e$ in $\mathrm{m}^{-3}$. Since ions and atoms have nearly equal masses and exchange energy efficiently, the length scale for the equilibration of the heavy-species temperature is $\lambda_{T_h} = 1/[(n_i + n_n)\overline{Q}_{in}]$. Here $\overline{Q}_{in} = 2[7 \times 10^{-10} - 0.6 \times 10^{-10} \ln(2k_B T_h/e)]^2$ m$^2$ is the ion–neutral momentum-transfer cross section [32].

At the smallest scale, the Debye length calculated from the LTE number densities, $\lambda_{De} = \sqrt{\varepsilon_0 k_B T_e/(n_e e^2)}$, is of order $10^{-8}$ m. It sets the extent over which charge separation and space-charge effects are important. These scales are reported in Table 1, together with electron mean free path $\lambda_e = 1/(n_n \overline{Q}_{en} + n_i \overline{Q}_{ei})$ and ion mean free path $\lambda_i = 1/(n_n \overline{Q}_{in} + n_i \overline{Q}_{ii})$, where $\overline{Q}_{ii}$ is the Coulomb cross section $\overline{Q}_{ei}$ evaluated at $T_h$.

The resulting hierarchy of length scales is $\lambda_{De} \ll \lambda_i \sim \lambda_e \sim \lambda_{T_h} < \lambda_{\mathrm{rxn}} \sim \lambda_T$. These scale estimates indicate that a continuum description is valid in the quasi-neutral plasma. In the sheath, whose thickness is a few tens of Debye lengths and thus comparable to $\lambda_i$ and $\lambda_e$, it is marginal; its validity there is examined a posteriori in Section 4.3. Furthermore, the gas temperature near the electrodes varies over a fluid boundary-layer thickness of order 1 mm, which is much larger than $\lambda_{T_h}$. Thus, the heavy species, consisting of ions and neutrals, can reasonably be described using a single temperature. These scales also indicate that the effect of electrodes is felt over a region of thickness of order $10^{-4}$ m by plasma, in agreement with other modelling results and analyses

**Table 1.** Characteristic scales of LTE argon at atmospheric pressure ($T_e = T_h = T$, Saha equilibrium), ordered from left to right by increasing size ($\lambda_i \approx \lambda_e$, and $\lambda_{rxn} \approx \lambda_T$ at 12 000 and 14 000 K). The reaction scales use the rate coefficients of Hoffert and Lien [33].

| $T$ (K) | $\lambda_{De}$ (m) | $\lambda_i$ (m) | $\lambda_e$ (m) | $\lambda_{T_h}$ (m) | $\lambda_{rxn}$ (m) | $\lambda_T$ (m) |
|---|---|---|---|---|---|---|
| 12 000 | $3.0 \times 10^{-8}$ | $8.1 \times 10^{-7}$ | $1.2 \times 10^{-6}$ | $2.1 \times 10^{-6}$ | $2.3 \times 10^{-4}$ | $2.3 \times 10^{-4}$ |
| 14 000 | $2.1 \times 10^{-8}$ | $6.6 \times 10^{-7}$ | $7.5 \times 10^{-7}$ | $3.2 \times 10^{-6}$ | $1.5 \times 10^{-4}$ | $1.4 \times 10^{-4}$ |
| 18 000 | $2.1 \times 10^{-8}$ | $8.8 \times 10^{-7}$ | $8.9 \times 10^{-7}$ | $5.9 \times 10^{-6}$ | $8.8 \times 10^{-5}$ | $1.7 \times 10^{-4}$ |
| 20 000 | $2.3 \times 10^{-8}$ | $1.1 \times 10^{-6}$ | $1.1 \times 10^{-6}$ | $6.9 \times 10^{-6}$ | $6.9 \times 10^{-5}$ | $2.2 \times 10^{-4}$ |
| 22 000 | $2.5 \times 10^{-8}$ | $1.4 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | $7.8 \times 10^{-6}$ | $5.6 \times 10^{-5}$ | $2.8 \times 10^{-4}$ |

[24, 19, 22, 20]. Consistent with this estimate, Lowke and Tanaka [16] showed that excluding a near-electrode region of this thickness is sufficient for the remaining plasma to be modelled as being in LTE. This separation of scales necessitates treating the near-electrode region separately from the bulk plasma.

We therefore model the electrode region as a one-dimensional domain of length $5 \times 10^{-4}$ m, with the electrode surface at one end and the bulk plasma at the other. Externally imposed bulk flows, such as the cathode jet, are not represented. The heavy gas is nevertheless not assumed to be at rest, since it can move during the transient stages of the discharge. The plasma is taken to consist of three species: electrons (e), singly charged argon ions (i), and argon atoms (n). Multiply charged ions are neglected: in the anode region the temperature is too low to produce them, and in the cathode region they are present but have a negligible effect [34]. Each species is assumed to follow a Maxwellian velocity distribution (see Appendix A).

Solutions are sought for the species number densities, the electron and heavy-particle temperatures, and the electric field. The following governing equations are thus solved in the electrode region.

The number densities of the three species obey

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \mathbf{\Gamma}_e = S, \qquad \frac{\partial n_i}{\partial t} + \nabla \cdot \mathbf{\Gamma}_i = S, \qquad \frac{\partial n_n}{\partial t} + \nabla \cdot \mathbf{\Gamma}_n = -S, \tag{1}$$

Different authors have evaluated the net ionization rate $S$ in different ways [24, 19, 16, 31]. In this work we use the reaction rates as proposed by Almeida et al. [24], where ionization is limited by the excitation of the first excited level, and accounts for the radiative escape of resonance radiation when $n_e \lesssim 10^{21}$ m$^{-3}$.

The momentum equation for the plasma as whole is given by $\rho\, D\mathbf{u}/Dt = \mathbf{F}$, where $\mathbf{F}$ is the net force on the plasma. For each species $s$, the corresponding momentum equation is $m_s n_s\, D_s\mathbf{u}_s/Dt = \mathbf{F}_s$, where $\mathbf{F}_s$ is the net force on the species. We neglect the inertia of each species relative to the heavy gas, $D_s\mathbf{u}_s/Dt \simeq D\mathbf{u}/Dt$, so that every species shares the acceleration of the heavy gas and there is no relative acceleration between species. The inertia of species $s$ is therefore $Y_s\mathbf{F}$, where $Y_s = m_s n_s/\rho$ is its mass fraction. The momentum equations then reduce to the local force balances [35]

$$0 = -en_e\mathbf{E} - \nabla p_e - m_e n_e \bar{\nu}^M_{en}(\mathbf{u}_e - \mathbf{u}_n) - m_e n_e \bar{\nu}^M_{ei}(\mathbf{u}_e - \mathbf{u}_i) - C_e n_e k_B \nabla T_e - Y_e\mathbf{F}, \tag{2}$$

$$0 = +en_i\mathbf{E} - \nabla p_i - Mn_i \bar{\nu}^M_{in}(\mathbf{u}_i - \mathbf{u}_n) - Mn_i \bar{\nu}^M_{ie}(\mathbf{u}_i - \mathbf{u}_e) + C_i n_e k_B \nabla T_e - Y_i\mathbf{F}, \tag{3}$$

$$0 = -\nabla p_n - Mn_n \bar{\nu}^M_{ni}(\mathbf{u}_n - \mathbf{u}_i) - Mn_n \bar{\nu}^M_{ne}(\mathbf{u}_n - \mathbf{u}_e) + C_n n_e k_B \nabla T_e - Y_n\mathbf{F}. \tag{4}$$

The frequencies $\bar{\nu}^M_{en}$, $\bar{\nu}^M_{ei}$, $\bar{\nu}^M_{in}$, $\bar{\nu}^M_{ie}$, $\bar{\nu}^M_{ni}$, and $\bar{\nu}^M_{ne}$ are *averaged momentum-transfer collision frequencies*. $C_e$ is the electron thermal coefficient, and $C_i$ and $C_n$ are its reactions on the ions and atoms. The calculations of these frequencies and coefficients are described in Appendix A. Momentum conservation in each collision pair relates the frequencies, and the thermal forces sum to zero:

$$m_e n_e \bar{\nu}^M_{en} = Mn_n \bar{\nu}^M_{ne}, \qquad m_e n_e \bar{\nu}^M_{ei} = Mn_i \bar{\nu}^M_{ie}, \qquad n_i \bar{\nu}^M_{in} = n_n \bar{\nu}^M_{ni}, \qquad C_i + C_n = C_e \tag{5}$$

Adding equations (2)–(4) and using equation (5) results in the total momentum equation.

$$\rho\, D\mathbf{u}/Dt = e(n_i - n_e)\mathbf{E} - \nabla p. \tag{6}$$

In practice, equations (2), (3) and (6) are solved instead of (4), and the neutral velocity $\mathbf{u}_n$ is obtained $\mathbf{u}_n$ from the definition of heavy-gas velocity $\mathbf{u}$ as the mass-weighted average of the ion and neutral velocities,

$$\mathbf{\Gamma}_n = (n_i + n_n)\mathbf{u} - n_i\mathbf{u}_i. \tag{7}$$

Solving the electron balance (2) for the electron flux, with $\bar{\nu}_e^M = \bar{\nu}_{en}^M + \bar{\nu}_{ei}^M$ the total electron momentum-transfer frequency and $\nabla p_e = k_B T_e \nabla n_e + n_e k_B \nabla T_e$, gives

$$\mathbf{\Gamma}_e = -\frac{e n_e}{m_e \bar{\nu}_e^M}\mathbf{E} - \frac{k_B T_e}{m_e \bar{\nu}_e^M}\nabla n_e - (1 + C_e)\,\frac{n_e k_B}{m_e \bar{\nu}_e^M}\nabla T_e + \frac{n_e}{\bar{\nu}_e^M}\Big(\bar{\nu}_{en}^M \mathbf{u}_n + \bar{\nu}_{ei}^M \mathbf{u}_i\Big), \tag{8}$$

where the negligible term $Y_e\mathbf{F}$ has been dropped. The first two terms are the conventional drift and diffusion. Their coefficients, $e/(m_e\bar{\nu}_e^M)$ and $k_B T_e/(m_e\bar{\nu}_e^M)$, are the magnitudes of the electron mobility and diffusivity; because both are set by the same frequency, they satisfy the Einstein relation identically. The third term combines the temperature-gradient part of the electron pressure force with the thermal force. The last term is the collisional drag exerted on the electrons by the moving heavy species.

The ion flux follows from the ion balance (3) in the same way. With $\bar{\nu}_i^M = \bar{\nu}_{in}^M + \bar{\nu}_{ie}^M$ the total ion momentum-transfer frequency and $\nabla p_i = k_B T_h \nabla n_i + n_i k_B \nabla T_h$,

$$\mathbf{\Gamma}_i = \frac{e n_i}{M \bar{\nu}_i^M}\mathbf{E} - \frac{k_B T_h}{M \bar{\nu}_i^M}\nabla n_i - \frac{n_i k_B}{M \bar{\nu}_i^M}\nabla T_h + C_i\,\frac{n_e k_B}{M \bar{\nu}_i^M}\nabla T_e - \frac{n_i}{n_h}\frac{\mathbf{F}}{M \bar{\nu}_i^M} + \frac{n_i}{\bar{\nu}_i^M}\Big(\bar{\nu}_{in}^M \mathbf{u}_n + \bar{\nu}_{ie}^M \mathbf{u}_e\Big). \tag{9}$$

The first three terms are ion drift, diffusion, and diffusion driven by the heavy-particle temperature gradient, with mobility $e/(M\bar{\nu}_i^M)$ and diffusivity $k_B T_h/(M\bar{\nu}_i^M)$. The fourth term is the reaction of the ions to the electron thermal force. The fifth term is the ions' share, $Y_i\mathbf{F}$, of the net force on the plasma. This share accelerates the ions together with the rest of the heavy gas. The last term is the collisional drag exerted on the ions by the atoms and electrons.

The velocities in the drag terms of equations (8) and (9) are those of the other species, so the two fluxes are coupled. If the drag, thermal-force, and mixture-acceleration terms are removed ($C_e = C_i = 0$, $Y_i\mathbf{F} = 0$, and $\mathbf{u}_s = \mathbf{u} = 0$ in the drag terms), equations (8) and (9) reduce to the classical drift–diffusion fluxes, with mobilities $e/(m_e\bar{\nu}_e^M)$ and $e/(M\bar{\nu}_i^M)$. Unlike the classical fluxes, however, they retain the full pressure-gradient force of each species in a non-isothermal plasma.

The electric field is obtained from Poisson's equation,

$$\nabla \cdot (\varepsilon_0 \mathbf{E}) = e(n_i - n_e), \qquad \mathbf{E} = -\nabla\phi, \tag{10}$$

which is solved over the entire computational domain. The transition from the quasi-neutral bulk plasma to the space-charge region is thereby captured without imposing the sheath boundaries a priori.

The electron thermal energy density $\mathcal{E} = \frac{3}{2} n_e k_B T_e$ obeys

$$\frac{\partial \mathcal{E}}{\partial t} + \nabla \cdot \mathbf{h}_e = -e\,\mathbf{\Gamma}_e \cdot \mathbf{E} - \frac{3}{2} n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^E - \varepsilon_{\text{ion}} S - w_{\text{rad}} - Q_e, \tag{11}$$

with the electron energy flux

$$\mathbf{h}_e = \frac{5}{2} k_B T_e\,\mathbf{\Gamma}_e - \kappa_e \nabla T_e + k_B T_e\, n_e \Big[C_i(\mathbf{u}_e - \mathbf{u}_i) + C_n(\mathbf{u}_e - \mathbf{u}_n)\Big]. \tag{12}$$

The terms on the right-hand side of equation (11) are Joule heating by the field, elastic energy transfer to the heavy species, the ionization-energy sink, the net radiative loss, and the radial heat loss of the electrons. In the elastic term, $\bar{\nu}_{eh}^E = \bar{\nu}_{en}^E + \bar{\nu}_{ei}^E$, where the superscript $E$ denotes the energy-exchange average of collision frequencies (Appendix A). The ionization sink is the net ionization rate multiplied by $\varepsilon_{\text{ion}} = 15.76$ eV, the first ionization potential of argon. The energy flux (12) is the sum of the enthalpy flux at $\frac{5}{2}k_B T_e$ per electron, Fourier conduction with the zero-current electron thermal conductivity $\kappa_e$, and a thermal-diffusion heat flux [35]. The net radiative loss $w_{\text{rad}}$ is that of Almeida et al. [24], taken from Benilov and Naidis [32].

A one-dimensional domain has no lateral boundary, so the Joule heat generated away from the electrode can leave only by radiation and by conduction through the two ends of the domain. In a real arc, the column loses most of its Joule heat radially, by conduction [36]. Without a lateral sink the one-dimensional column overheats, and the solution depends on the length of the domain. We therefore represent the radial conduction of a column of radius $R$ by a volumetric loss for each energy channel,

$$Q_e = \Lambda\,(\kappa_e + \kappa_R)\,(T_e - T_{\text{edge}}), \qquad Q_h = \Lambda\,\kappa_h\,(T_h - T_{\text{edge}}), \qquad \Lambda = \frac{4}{R^2}, \tag{13}$$

where $T_{\text{edge}}$ is the temperature at the edge of the column. Radially, ion pairs also diffuse out of the hot core and recombine near the cold edge, releasing the ionization energy there. The axial

counterpart of this transport is carried by the model's own equations, through the ambipolar flux and the ionization sink (Appendix A), but radially it has to be represented by the reactive thermal conductivity $\kappa_R$. We take $\kappa_R$ of argon in LTE at atmospheric pressure computed with Mutation++ [37], the same as in figure 12, evaluated at $T_e$. It is charged to the electrons, because the ionization energy is electron energy in this model. The factor $\Lambda$ follows from a parabolic radial temperature profile, for which $-r^{-1}\,\partial_r(r\kappa\,\partial_r T) = 4\kappa(T - T_{\mathrm{edge}})/R^2$ on the axis. Each channel loses heat through its own conductivity, so the gas is not cooled through the electrons, and $T_e - T_h$ remains set by the elastic exchange. $Q_e$ enters the electron energy equation (11) and $Q_h$ the heavy-particle energy equation (15). The radius is not prescribed; $\Lambda$ is instead chosen so that the bulk-plasma state imposed at $x = L$ (Section 2.2) is a uniform solution of the column, in which Joule heating balances radiation and radial conduction,

$$\frac{J_{\mathrm{app}}^2}{\sigma} = w_{\mathrm{rad}} + \Lambda\,(\kappa_e + \kappa_R + \kappa_h)\,(T_{\mathrm{bulk}} - T_{\mathrm{edge}}), \tag{14}$$

with all quantities evaluated at the bulk state, and $\sigma$ the electrical conductivity of the plasma. The plasma far from the electrode then relaxes to the bulk state, whatever the length of the domain. Within the near-electrode layer, the axial gradients develop over $10^{-6}$–$10^{-4}$ m, while $R$ is of order $10^{-3}$ m. There, the radial losses are negligible in comparison.

The common heavy-particle temperature is governed by

$$n_h c_v \left(\frac{\partial T_h}{\partial t} + \mathbf{u}\cdot\nabla T_h\right) = \nabla\cdot(\kappa_h \nabla T_h) + \frac{3}{2} n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^{E} + Q_{in} - Q_h - n_h k_B T_h\,\nabla\cdot\mathbf{u}, \tag{15}$$

where $c_v = \frac{3}{2}k_B$ is the translational heat capacity per heavy particle and $\kappa_h$ is the heavy-particle thermal conductivity (Appendix A). The elastic-exchange term is the same as in the electron energy equation (11), but with the opposite sign. The ionization energy and the radiative loss are charged entirely to the electrons and do not appear here. $Q_h$ is the radial heat loss of the heavy particles, equation (13). The ion–neutral frictional heating is

$$Q_{in} = M n_i\,\bar{\nu}_{in}^{M}\,|\mathbf{u}_i - \mathbf{u}_n|^2, \tag{16}$$

which is positive definite and reduces to the Joule dissipation of the ion drift when the neutral gas is at rest. The last term in equation (15) is the compression work of the heavy gas.

The heavy-gas velocity is advanced by an inviscid single-fluid momentum equation,

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\mathbf{u}\mathbf{u}) = -\nabla p + e(n_i - n_e)\mathbf{E}, \qquad p = n_e k_B T_e + n_h k_B T_h. \tag{17}$$

This equation governs the motion of the heavy gas as a whole.

The state advanced in time is therefore $\{n_e, n_i, n_n, T_e, T_h, \mathbf{u}, \phi\}$. It is governed by the continuity equations (1), Poisson's equation (10); the electron and heavy-particle energy equations (11), (15) and the heavy-gas momentum equation (17).

### *2.2 Boundary conditions*

The system is closed by conditions at the two ends of the domain: the electrode surface, $x = 0$, and the edge of the near-electrode layer, $x = L$, where the solution is matched to the adjacent LTE bulk plasma. The potential is referenced to zero at the electrode, $\phi(0) = 0$. Electrons reaching the surface are absorbed; they are assumed to have a Maxwellian distribution at their local temperature $T_e$. At the cathode, thermionic emission of electrons, $\Gamma_{\mathrm{em}}$, with a Maxwellian distribution at the wall temperature $T_{\mathrm{wall}}$, is also included. The electron flux into the surface is

$$\boldsymbol{\Gamma}_e\cdot\hat{\mathbf{n}} = \tfrac{1}{4} n_e \bar{c}_e - \Gamma_{\mathrm{em}}, \qquad \bar{c}_e = \sqrt{\frac{8k_B T_e}{\pi m_e}}. \tag{18}$$

The emitted flux is given by the Richardson–Dushman law with Schottky lowering of the work function $W$ by the field at the surface,

$$\Gamma_{\mathrm{em}} = \frac{A_R T_{\mathrm{wall}}^2}{e}\exp\!\left(-\frac{W - \Delta W}{k_B T_{\mathrm{wall}}}\right), \qquad \Delta W = \sqrt{\frac{e^3 E_w}{4\pi\varepsilon_0}}, \qquad E_w = \max(\mathbf{E}\cdot\hat{\mathbf{n}}, 0), \tag{19}$$

where $A_R$ is the Richardson constant. Each absorbed electron removes a mean energy $2k_B T_e$ from the plasma, and each emitted electron brings in $2k_B T_{\mathrm{wall}}$, so the boundary condition for the electron energy equation (11) is

$$\mathbf{h}_e\cdot\hat{\mathbf{n}} = 2k_B T_e\,\tfrac{1}{4} n_e \bar{c}_e - 2k_B T_{\mathrm{wall}}\,\Gamma_{\mathrm{em}}. \tag{20}$$

Let $\mathbf{\Gamma}_i^d$ denote the ion flux (9) without its density-diffusion term, $\mathbf{\Gamma}_i^d = \mathbf{\Gamma}_i + [k_B T_h/(M\bar{\nu}_i^M)]\nabla n_i$, which contains the field drift and the thermal, thermal-diffusion, mixture-force, and drag contributions. Ions are absorbed at their thermal-effusion flux, plus this directed flux where it points into the wall,

$$\mathbf{\Gamma}_i \cdot \hat{\mathbf{n}} = \tfrac{1}{4} n_i \bar{c}_i + \max\left(\mathbf{\Gamma}_i^d \cdot \hat{\mathbf{n}}, 0\right), \qquad \bar{c}_i = \sqrt{\frac{8 k_B T_{\text{wall}}}{\pi M}}, \tag{21}$$

which is similar to the form of Hagelaar et al. [38]. Where the sheath field drives the ions into the electrode, the directed term dominates. Elsewhere, the thermal-effusion flux ensures that ions reaching the wall are still absorbed where transport alone does not carry them into it.

The wall is impermeable to the heavy gas, $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$, which is the boundary condition for the heavy-gas momentum equation (17), and the pressure at the electrode surface is taken to be atmospheric pressure. Equation (7) then gives

$$\mathbf{\Gamma}_n \cdot \hat{\mathbf{n}} = -\mathbf{\Gamma}_i \cdot \hat{\mathbf{n}} : \tag{22}$$

every ion neutralized at the surface re-enters the plasma as an atom. Secondary electron emission by ion impact is not included.

The heavy-particle temperature at the surface equals the electrode temperature, $T_h(0) = T_{\text{wall}}$. Unless it is prescribed, $T_{\text{wall}}$ is obtained at every time step by balancing the heat flux delivered by the plasma against steady conduction through the electrode body,

$$q_w = \frac{k_w}{L_w}\left(T_{\text{wall}} - T_0\right), \tag{23}$$

where $k_w$ is the thermal conductivity of the electrode, $L_w$ its length, and $T_0$ the temperature of its cooled base. The heat flux into the surface is

$$q_w = \left(2k_B T_e + W\right)\tfrac{1}{4} n_e \bar{c}_e - \left(2k_B T_{\text{wall}} + W\right)\Gamma_{\text{em}} + \left(\varepsilon_{\text{ion}} - W\right)\mathbf{\Gamma}_i \cdot \hat{\mathbf{n}} - \kappa_h \nabla T_h \cdot \hat{\mathbf{n}}. \tag{24}$$

The first term is the energy delivered by absorbed electrons, their kinetic energy plus the work function released on entering the metal. The second is emission cooling: each emitted electron removes the work function and its kinetic energy. The third is the energy released when an ion recombines at the surface, $\varepsilon_{\text{ion}}$, less the work function spent to extract the neutralizing electron. The last is heat conduction from the heavy gas. At the anode, $\Gamma_{\text{em}} = 0$ and the emission term vanishes. The heat flux $q_w$ decreases as $T_{\text{wall}}$ increases, steeply through emission cooling at the cathode and through the gas conduction at both electrodes, apart from the weak $T_{\text{wall}}^{1/2}$ dependence of the ion effusion flux. The right-hand side of equation (23) rises linearly, so the balance has a single root, which is found by bisection.

*Bulk-plasma edge, $x = L$.* The electron and ion densities and both temperatures are set to Dirichlet values from the adjacent LTE bulk plasma: $n_e = n_e^{\text{bulk}}$, $n_i = n_i^{\text{bulk}}$, and $T_e = T_h = T_{\text{bulk}}$; the last of these is the boundary condition for both energy equations. This bulk state, together with the applied current density $J_{\text{app}}$, also fixes the radial-loss coefficient $\Lambda$ through equation (14), so that the imposed values are consistent with the energy balance of the column. The neutral density follows from Dalton's law at the background pressure $p_0$,

$$n_n = \frac{p_0 - n_e k_B T_e}{k_B T_h} - n_i. \tag{25}$$

This edge is open to the heavy gas: the heavy-gas momentum equation (17) sees the bulk state, and hence the pressure $p_0$, outside the boundary, and the heavy-gas velocity is given a zero normal gradient. The arc current is imposed through the electric field in Poisson's equation (10): the total current density at the boundary, conduction plus displacement, is held equal to the applied current density $J_{\text{app}}$,

$$e\left(\mathbf{\Gamma}_i - \mathbf{\Gamma}_e\right) + \varepsilon_0 \frac{\partial E}{\partial t} = J_{\text{app}}, \tag{26}$$

which is integrated in time as an evolution equation for the boundary field. The initial field enters only as the initial condition of this equation.

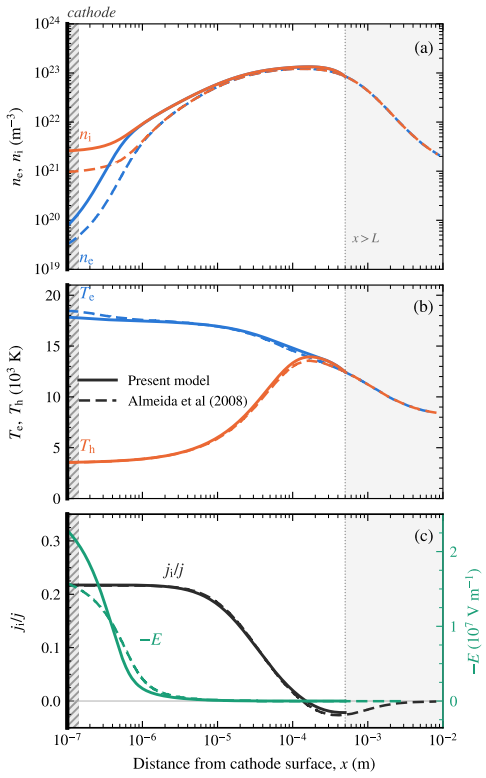



**Figure 1.** Near-cathode layer in argon at atmospheric pressure, cathode surface temperature $T_w = 3500$ K and current density $j_c = 10^7$ A m$^{-2}$: present model (solid lines) and Almeida et al. [24] (dashed lines, figures 5(b) and 6). (a) Electron and ion number densities. (b) Electron and heavy-particle temperatures. (c) Ion fraction of the current density, $j_i/j$ (left axis), and field strength $-E$ (right axis). The hatched strip at the left marks the cathode surface. The shaded region, $x > L$, lies outside the computational domain.

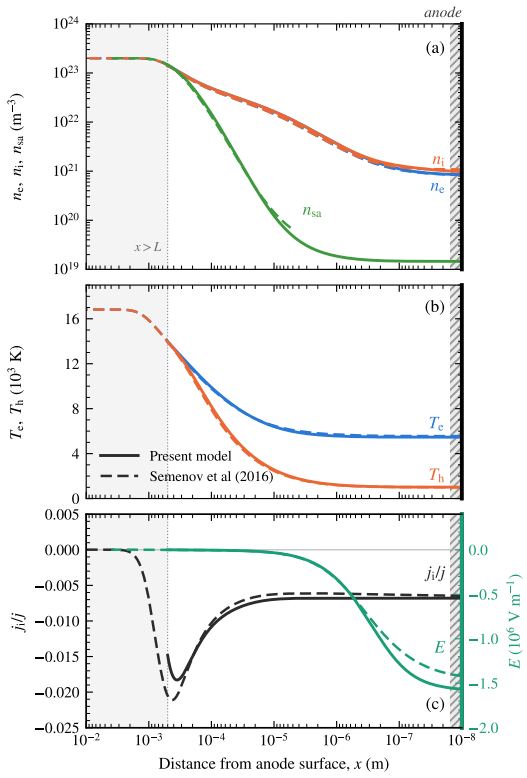



**Figure 2.** Near-anode layer in argon at atmospheric pressure, anode surface temperature $T_s = 1000$ K and current density $j = 1.5 \times 10^7$ A m$^{-2}$: present model (solid lines) and Semenov et al. [26] (dashed lines, figures 3(b), 4 and 5). (a) Electron and ion number densities and the Saha density $n_{sa}$. (b) Electron and heavy-particle temperatures. (c) Ion fraction of the current density, $j_i/j$ (left axis), and electric field $E$ (right axis). The distance axis is reversed, so that the anode surface, marked by the hatched strip, is at the right. The shaded region, $x > L$, lies outside the computational domain.

## 3 Validation

The model is validated against two published unified models of the near-electrode layers of atmospheric-pressure argon arcs: the near-cathode layer computed by Almeida et al. [24] and the near-anode layer computed by Semenov et al. [26]. Both works solve the multi-fluid equations for electrons, ions and atoms with separate electron and heavy-particle temperatures, coupled to Poisson's equation, across the whole layer, from the electrode surface through the space-charge sheath and the ionization layer to the bulk plasma. Neither assumes quasi-neutrality or matches separate sheath and plasma solutions, so complete profiles can be compared with those of the present model. The reference profiles were digitized from the published figures.

The conditions of the reference calculations were reproduced as closely as possible. The electrode surface temperature was prescribed at the value used in each reference, rather than obtained from the heat balance (23), and the radial loss (13) was switched off, $\Lambda = 0$, since neither reference includes it. The domain length was $L = 5 \times 10^{-4}$ m in both cases, and the Dirichlet state imposed at $x = L$ (Section 2.2) was taken from the reference profiles at that distance, where $T_e \approx T_h$ in both. The domain was discretized with a uniform base grid of 512 cells and four levels of refinement towards the electrode, each with a refinement ratio of 4, so that the finest cell size is $3.8 \times 10^{-9}$ m, several times smaller than the Debye length near the electrode (Table 1). The equations were integrated in time at a fixed current density until a steady state was reached. In both cases $x$ denotes the distance from the electrode surface.

### *3.1 Near-cathode layer*

The cathode case is that of figure 5(b) of Almeida et al. [24]: a tungsten cathode with work function 4.55 eV, thermionic emission (19) included, surface temperature 3500 K and current

density $10^7$ A m$^{-2}$. The two solutions are compared in figure 1. The temperature profiles agree closely across the whole layer (figure 1(b)). The electron temperature stays near 17.5–18 kK through the sheath and the inner part of the ionization layer and decreases towards the bulk plasma, while the heavy-particle temperature rises from the wall value to a maximum of 13.9 kK at $x \approx 2 \times 10^{-4}$ m, compared with 13.6 kK in the reference. From $x \approx 3 \times 10^{-6}$ m outwards, both temperatures agree with the reference within 0.4 kK (3 %).

The ion fraction of the current, figure 1(c), is reproduced over the whole layer. At the cathode surface the present model gives $j_i/j = 0.218$, compared with 0.216 reported by Almeida et al.; $j_i/j$ stays at this value through the sheath, falls to zero at $x \approx 1.5 \times 10^{-4}$ m, and becomes slightly negative further out, where the ion flux is directed away from the cathode. The maximum potential in the layer is 13.6 V, compared with 13.95 V in the reference. In the ionization layer, $10^{-5} \lesssim x \lesssim 3 \times 10^{-4}$ m, the charged-particle densities of the present model are 10–15 % higher (figure 1(a)).

The largest differences are in the space-charge sheath, $x \lesssim 10^{-6}$ m. At $x = 10^{-7}$ m, the electron and ion densities of the present model are higher by a factor of about 2.5, and the field strength is $2.3 \times 10^7$ V m$^{-1}$, compared with $1.6 \times 10^7$ V m$^{-1}$. Since the ion current fraction is the same in both models, the higher ion density in the sheath corresponds to a lower ion drift velocity. The sheath is where the solution is most sensitive to the description of ion transport at a high reduced field (Appendix A) and to the ion boundary condition (21). These differences remain confined to the sheath and its edge: further out, the field is small in both models, and the current fractions and the temperatures agree.

### *3.2 Near-anode layer*

The anode case is that of figure 3(b) of Semenov et al. [26]: a non-emitting anode, $\Gamma_{\mathrm{em}} = 0$, with surface temperature 1000 K and current density $1.5 \times 10^7$ A m$^{-2}$ (1500 A cm$^{-2}$). The comparison is shown in figure 2. The electron and ion densities agree within 15 % across the layer (figure 2(a)). Following Semenov et al., the figure also shows the Saha density $n_{\mathrm{sa}}$, the charged-particle density that the local $T_e$ and $T_h$ would give in ionization equilibrium with quasi-neutrality at the pressure $p_0$. Near the anode the actual densities exceed $n_{\mathrm{sa}}$ by nearly two orders of magnitude. The anode region is therefore far from ionization equilibrium: charged particles are supplied by diffusion from the hotter plasma rather than produced locally, and both models capture this departure.

The temperature profiles agree closely (figure 2(b)). At the anode surface, the electron temperature is 5.45 kK, compared with 5.59 kK reported by Semenov et al., and the heavy-particle temperatures differ by less than 0.3 kK across the layer. The electric field near the anode, figure 2(c), is negative: it is directed towards the anode, so that the anode lies at a lower potential than the adjacent plasma and the electrons are retarded. The field at the surface is $-1.57 \times 10^6$ V m$^{-1}$, compared with $-1.42 \times 10^6$ V m$^{-1}$ in the reference. Ions flow into the anode, so $j_i/j$ is negative there. At the surface, the present model gives $j_i/j = -0.0068$, compared with $-0.0065$ in the reference.

The main difference is in the minimum of $j_i/j$, which is $-0.018$ at $x = 3.5 \times 10^{-4}$ m in the present model and $-0.021$ at $x = 4.3 \times 10^{-4}$ m in the reference. This minimum lies close to the domain boundary $x = L$. In the reference solution, the plasma becomes uniform only at $x \approx 1$–$2$ mm, where $T_e = 16.8$ kK, whereas the present calculation imposes the reference state at $x = L = 0.5$ mm, where $T_e = T_h \approx 14$ kK and the plasma is still varying. The Dirichlet condition at $x = L$ therefore constrains the solution in the outer part of the ionization layer, which is the likely cause of the difference near $x = L$. Closer to the anode, the two solutions agree.

In both cases, the present model reproduces the structure of the near-electrode layer obtained by independent unified models: the temperatures, the current partition between ions and electrons, the sign and magnitude of the near-electrode field, and the departure from ionization equilibrium. Apart from the charged-particle densities in the cathode sheath and the minimum of $j_i/j$ near the outer boundary of the anode calculation, the temperatures agree within 3–5 % and the charged-particle densities within 15 %.

## 4 Results

The model is now applied to an anode and a cathode of a high-intensity arc in argon at atmospheric pressure. The two base cases differ in the bulk-plasma state at $x = L$, in the current density and in electron emission (table 2). Unlike the validation cases of Section 3, the radial loss (13) is included, with the reactive conductivity in the electron channel. The calibration (14) gives a column radius $R = 2.3$ mm for the anode and 0.42 mm for the cathode. At the anode bulk temperature, $\kappa_R = 0.59$ W m$^{-1}$ K$^{-1}$ is comparable to $\kappa_e = 0.76$ W m$^{-1}$ K$^{-1}$, whereas at 21 000 K

the plasma is almost fully ionized and $\kappa_R = 0.03\ \mathrm{W\,m^{-1}\,K^{-1}}$ is negligible. At $x = L$ all densities and temperatures are Dirichlet values (Section 2.2). The anode surface temperature is prescribed, $T_{\mathrm{wall}} = 1000$ K. The cathode surface temperature is not prescribed: it follows from the heat balance (23), with $k_w = 100\ \mathrm{W\,m^{-1}\,K^{-1}}$, $L_w = 1$ cm and $T_0 = 1000$ K. The electrode material has $W = 2.63$ eV and $A_R = 3 \times 10^4\ \mathrm{A\,m^{-2}\,K^{-2}}$. The domain, grid and refinement are those of Section 3. Each case was integrated in time at fixed current density until the relative change of every variable over 1 $\mu$s fell below $10^{-3}$. In both steady states, the sum of the terms of the electron energy equation (11), i.e. $\partial\mathcal{E}/\partial t$, is below $1.2 \times 10^{-3}$ of its largest term everywhere, and the total current density is uniform to within $10^{-8}$.

**Table 2.** Base cases: conditions (top) and computed electrode quantities (bottom). The sheath edge $x_{\mathrm{s}}$ is the distance from the electrode at which $|n_i - n_e|/n_i$ falls to 1 %, and $\phi_{\mathrm{s}}$ is the potential there; $\phi(L)$ is the potential at the bulk-plasma boundary. Both potentials are relative to the electrode. Fluxes are towards the electrode.

| | Anode | Cathode |
|---|---|---|
| Current density $\lvert J_{\mathrm{app}}\rvert$ ($\mathrm{A\,m^{-2}}$) | $9 \times 10^6$ | $1.2 \times 10^8$ |
| Bulk temperature $T_{\mathrm{bulk}}$ (K) | 12 400 | 21 000 |
| Bulk electron density $n_e^{\mathrm{bulk}}$ ($\mathrm{m^{-3}}$) | $8.2 \times 10^{22}$ | $1.7 \times 10^{23}$ |
| Thermionic emission | off | on |
| Surface temperature $T_{\mathrm{wall}}$ (K) | 1000 (prescribed) | 3698 |
| $T_e$ at the surface (K) | 4900 | 19 500 |
| $n_e$, $n_i$ at the surface ($\mathrm{m^{-3}}$) | $5.2 \times 10^{20}$, $4.7 \times 10^{20}$ | $2.1 \times 10^{21}$, $7.6 \times 10^{21}$ |
| Sheath edge, $x_{\mathrm{s}}$ (m) | $8.4 \times 10^{-7}$ | $5.7 \times 10^{-7}$ |
| $\phi_{\mathrm{s}}$, $\phi(L)$ (V) | 0.41, 2.60 | 4.13, 13.68 |
| Field at the surface ($\mathrm{V\,m^{-1}}$) | $-7.2 \times 10^5$ | $2.2 \times 10^7$ |
| $j_i/j$ at the surface | −0.0037 | 0.057 |
| Absorbed electron flux $\frac{1}{4} n_e \bar{c}_e$ ($\mathrm{m^{-2}\,s^{-1}}$) | $5.6 \times 10^{25}$ | $4.6 \times 10^{26}$ |
| Emitted electron flux $\Gamma_{\mathrm{em}}$ ($\mathrm{m^{-2}\,s^{-1}}$) | – | $1.17 \times 10^{27}$ |
| Ion flux $\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}}$ ($\mathrm{m^{-2}\,s^{-1}}$) | $2.1 \times 10^{23}$ | $4.3 \times 10^{25}$ |
| Heat flux $q_w$ ($\mathrm{MW\,m^{-2}}$) | 38.2 | 27.0 |

### *4.1 Structure of the near-electrode layers*

Figure 3 shows the steady profiles of both cases. In both, the plasma departs from LTE over a layer of order $10^{-4}$ m, as estimated in Section 2.1, but the structure of this layer is very different at the two electrodes.

*Anode.* The space-charge sheath is practically absent. Charge separation exceeds 1 % only within 0.84 $\mu$m of the surface, where the potential differs from that of the anode by 0.41 V, and at the surface $n_e$ exceeds $n_i$ by 9 % (figure 3(b)). The anode lies 2.60 V below the bulk-plasma boundary, and this drop is distributed over the whole quasi-neutral layer (figure 3(f)): 1.14 V within 10 $\mu$m, 1.83 V within 100 $\mu$m. The field is directed towards the anode and retards the electrons, which reach the surface by diffusion down the steep density gradient: the charged-particle density falls by a factor of 160, from $8.2 \times 10^{22}$ to $5.2 \times 10^{20}$ $\mathrm{m^{-3}}$, across the layer. The electron and heavy-particle temperatures decrease together towards the anode, but $T_h$ falls faster, to the prescribed surface temperature of 1 kK, so that the two separate progressively: by 2.1 kK at $x = 10^{-4}$ m, 3.6 kK at $x = 10^{-5}$ m, and 3.9 kK at the surface, where $T_e = 4.9$ kK (figure 3(d)). Ions flow into the anode, $j_i/j < 0$, and carry only 0.4 % of the current at the surface (figure 3(h)). The field at the anode, $7.2 \times 10^5\ \mathrm{V\,m^{-1}}$, is too weak to heat the ions: $T_{\mathrm{eff}}$ exceeds $T_h$ by about 0.1 kK.

*Cathode.* A space-charge sheath forms, 0.57 $\mu$m thick, in which $n_i$ exceeds $n_e$ by up to a factor of 3.6 at the surface (figure 3(a)), and across which the potential rises by 4.1 V. The field at the surface is $2.2 \times 10^7\ \mathrm{V\,m^{-1}}$. Beyond the sheath, the potential keeps rising through the quasi-neutral ionization layer, to 8.8 V at $x = 10^{-4}$ m (figure 3(e)). Further out, the plasma approaches the bulk state, and the remaining rise to 13.7 V at $x = L$ is mostly the ohmic drop of the column, whose field is $1.1 \times 10^4\ \mathrm{V\,m^{-1}}$. The electron temperature is nearly uniform, 19.5–21 kK, across the whole layer, while the heavy-particle temperature rises from the surface value of 3.7 kK and approaches $T_e$ only at $x \approx 10^{-4}$ m (figure 3(c)). Over the same distance the atom density falls by three orders of magnitude, from $2 \times 10^{24}$ to $2.6 \times 10^{22}$ $\mathrm{m^{-3}}$, and the plasma becomes almost fully ionized. The emitted electron flux, $1.17 \times 10^{27}\ \mathrm{m^{-2}\,s^{-1}}$, corresponds to a current density of $1.9 \times 10^8\ \mathrm{A\,m^{-2}}$, larger than the arc current density. It is partly returned by the plasma electrons that reach the surface against the sheath field, $4.6 \times 10^{26}\ \mathrm{m^{-2}\,s^{-1}}$ (table 2), and the ions carry 5.7 % of the current at the surface (figure 3(g)). In the sheath, the ions are strongly heated by the field: $T_{\mathrm{eff}}$ reaches 48 kK at the surface (figure 3(c)).

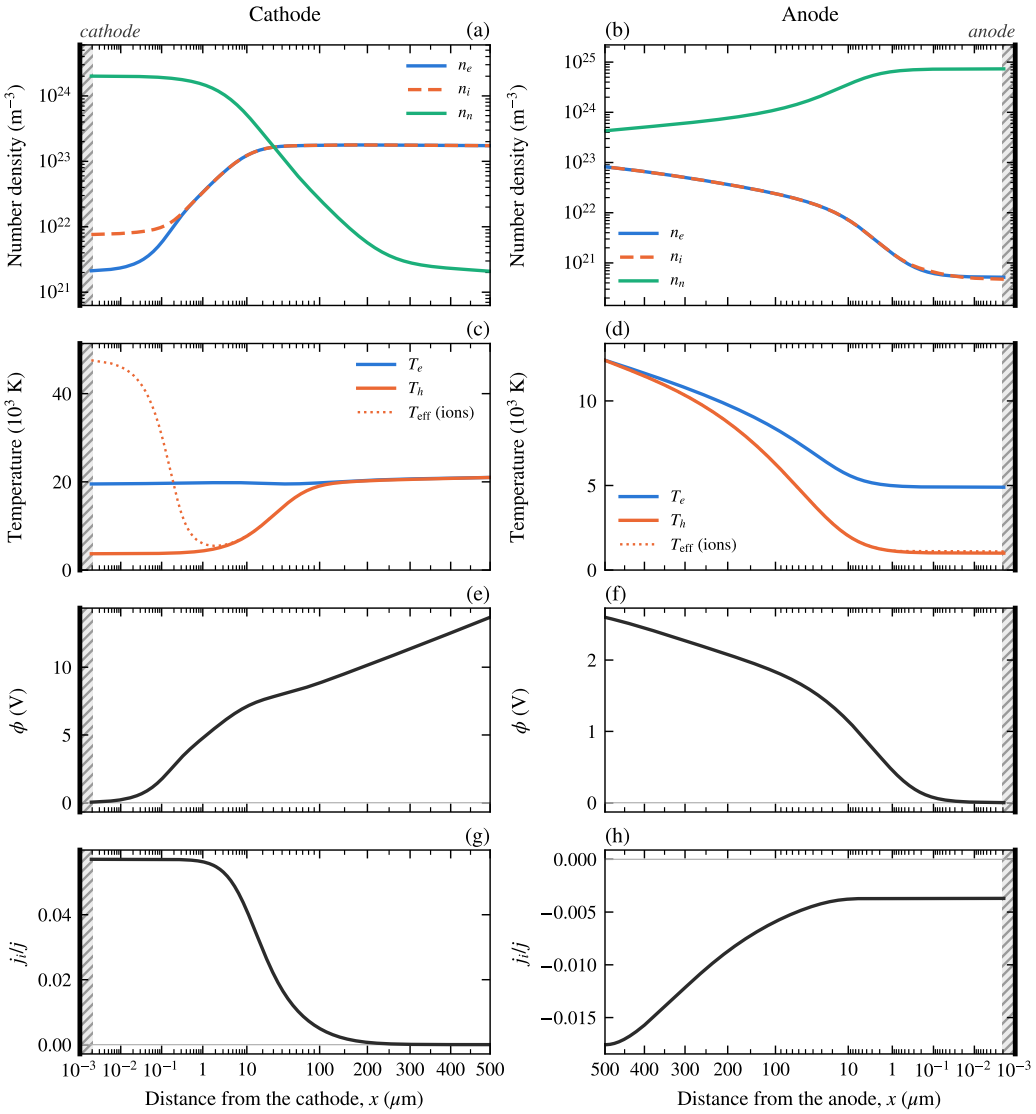



**Figure 3.** Base cases: cathode (left) and anode (right). $x$ is the distance from the electrode; the cathode surface is at the left edge of its panels and the anode surface at the right edge of its panels, marked by the hatched strip. The distance axis is scaled as $\ln x + x/x_c$, with $x_c = 50$ $\mu$m: logarithmic near the electrode and smoothly becoming linear towards the bulk plasma, so that both the sheath and the approach to the bulk are visible. (a), (b) Electron, ion and atom number densities. (c), (d) Electron and heavy-particle temperatures, and the effective ion temperature $T_{\rm eff}$ of the two-temperature ion mobility (Appendix A), shown where $n_n > n_i$, the range in which the ion–atom closure applies. (e), (f) Potential relative to the electrode. (g), (h) Ion fraction of the current density, positive for ions flowing into the electrode.

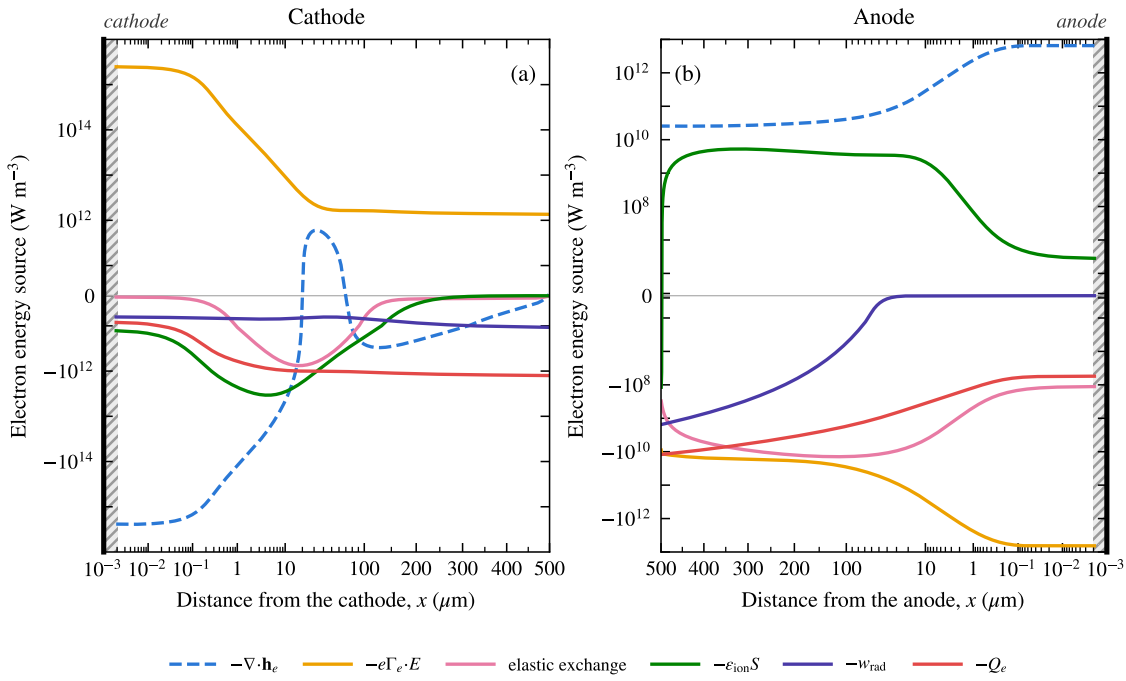



**Figure 4.** Steady electron energy balance, equation (11), for (a) the cathode and (b) the anode: every term except $\partial\mathcal{E}/\partial t$. Dashed: the divergence of the electron energy flux (12), $-\nabla\cdot\mathbf{h}_e$. Solid: the volume terms, i.e. Joule heating, elastic exchange with the heavy particles, the ionization sink and the radiative loss. Positive terms heat the electrons. The hatched strip marks the electrode surface: the cathode at the left of panel (a), the anode at the right of panel (b). The distance axis is scaled as $\ln x + x/x_c$, with $x_c = 50$ $\mu$m: logarithmic near the electrode and smoothly becoming linear towards the bulk plasma, so that both the sheath and the approach to the bulk are visible. The vertical axis is logarithmic in $|\cdot|$ above $10^{11}$ (a) and $10^{6}$ W m$^{-3}$ (b), and linear below.

### *4.2 Electron energy balance*

Figure 4 compares the terms of the electron energy equation (11) across the layer. The ordering of the terms is again opposite at the two electrodes.

*Anode.* The Joule term $-e\mathbf{\Gamma}_e\cdot\mathbf{E}$ is negative throughout the layer (figure 4(b)): the electrons flow towards the anode against the retarding field and lose energy to it, 23.5 MW m$^{-2}$ over the domain. This loss, and the smaller elastic (4.5 MW m$^{-2}$) and radial (2.6 MW m$^{-2}$) losses, are supplied by the electron energy flux from the column. At $x = L$, 36.4 MW m$^{-2}$ flow towards the anode, two thirds of it as the enthalpy flux $\frac{5}{2}k_BT_e\mathbf{\Gamma}_e$, and only 7.6 MW m$^{-2}$ reach the surface. The convergence of the enthalpy flux, as $T_e$ falls towards the anode, supplies 15.0 MW m$^{-2}$, conduction 4.2 MW m$^{-2}$ and the thermal-diffusion heat flux 9.6 MW m$^{-2}$. Close to the anode, $x < 10$ $\mu$m, conduction and thermal diffusion provide almost all of the supply, and within 0.1 $\mu$m of the surface the thermal-diffusion term alone balances the Joule loss. The volume exchange terms are smaller than the flux terms by two to three orders of magnitude near the anode, and become comparable to them only at $x \gtrsim 10^{-4}$ m, where the elastic loss to the heavy particles keeps $T_h$ close to $T_e$. The ionization term is positive over most of the layer: near the anode the plasma recombines, $S < 0$, because the charged particles are supplied by diffusion at densities well above the local ionization equilibrium (Section 3.2), and the recombination energy heats the electrons.

*Cathode.* Joule heating is the dominant electron energy source everywhere in the layer (figure 4(a)). It reaches $2.5\times10^{15}$ W m$^{-3}$ at the surface, where the emitted electrons are accelerated by the sheath field. 467 MW m$^{-2}$ are deposited in the sheath, and 1590 MW m$^{-2}$ over the whole domain. In the sheath, the Joule heating is balanced by the divergence of the conduction and thermal-diffusion fluxes (334 and 113 MW m$^{-2}$). The electron temperature is slightly lower at the surface than at the sheath edge, 19.5 against 19.7 kK, so these fluxes carry heat towards the cathode. The enthalpy flux of the electron current, directed away from the cathode, is large, 480–540 MW m$^{-2}$, but nearly uniform, and its divergence is an order of magnitude smaller than that of conduction. Conduction changes sign at $x \approx 1.4\times10^{-5}$ m and deposits energy in the ionization layer, out to $x \approx 10^{-4}$ m, where ionization (98 MW m$^{-2}$ within 100 $\mu$m) and the elastic heating of the heavy particles take it up. Over the whole domain, the Joule heat is distributed as follows: 690 MW m$^{-2}$ leave at $x = L$ as electron energy flux into the column (543 MW m$^{-2}$ of

enthalpy and 153 MW m$^{-2}$ of thermal-diffusion heat flux); 574 MW m$^{-2}$ are radial loss, most of it in the column part of the domain, $x > 10^{-4}$ m; 130 MW m$^{-2}$ reach the cathode with the absorbed plasma electrons, net of the energy of the emitted electrons; ionization takes 108 MW m$^{-2}$; and radiation and elastic exchange take 46 and 44 MW m$^{-2}$.

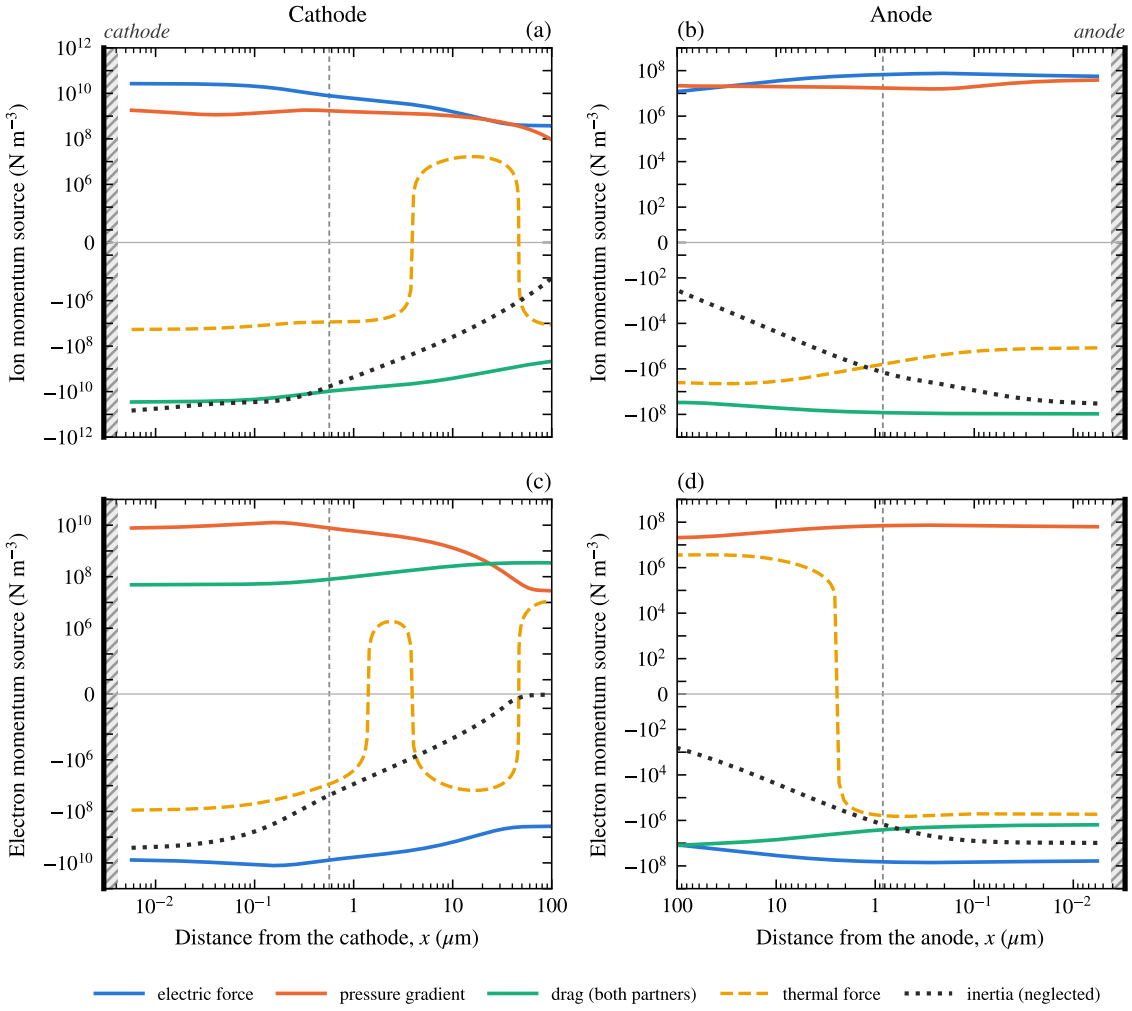



**Figure 5.** Terms of the steady momentum balances of the ions, equation (3), (a) and (b), and of the electrons, equation (2), (c) and (d), at the cathode (left) and the anode (right), within 100 $\mu$m of the electrode. Positive terms push the species towards the electrode. Drag: the friction with both collision partners combined; thermal force: the term in $C_e$ of equation (2) and its reaction $C_i$ on the ions. Dotted: the inertia $-m_s n_s(\mathbf{u}_s \cdot \nabla)\mathbf{u}_s$, which the model neglects, evaluated with $\mathbf{u}_s = \mathbf{\Gamma}_s/n_s$ from the computed fluxes. The vertical dashed line marks the sheath edge $x_s$ (table 2). The vertical axis is logarithmic in $|\cdot|$ and linear within six decades of the largest term.

### *4.3 Momentum balance and collisionality of the sheaths*

The fluxes (8) and (9) follow from the force balances (2) and (3), in which the inertia of each species relative to the heavy gas is neglected. Figure 5 shows the terms of these balances in the two steady states near the electrode: the electric force, the pressure gradient, the two collisional drags combined, and the thermal force. In the steady states the heavy gas is at rest ($|\mathbf{u}| \leq 4 \times 10^{-4}$ m s$^{-1}$ at the cathode and 0.16 m s$^{-1}$ at the anode), so the share $Y_s\mathbf{F}$ of the net force is negligible (below $2 \times 10^{-4}$ of the largest term). For comparison, the figure also shows the inertia $m_s n_s(\mathbf{u}_s \cdot \nabla)\mathbf{u}_s$ that the model neglects, evaluated a posteriori from the computed fluxes. The retained terms sum to zero to within $3 \times 10^{-3}$ of the largest one.

*Ions.* In both sheaths, the electric force on the ions is balanced by the drag (figure 5(a), (b)). At the cathode the pressure gradient adds at most 22 % to the electric force in the sheath, and 7 % at the surface. At the anode the field is weaker, and the pressure gradient, which also drives the ions towards the surface, is 68 % of the electric force there. The thermal force on the ions is negligible, below 1 % of the drag in either sheath. The neglected inertia is not small inside the cathode sheath. It exceeds the drag within 0.36 $\mu$m of the surface, by a factor of 2.4 at the surface,

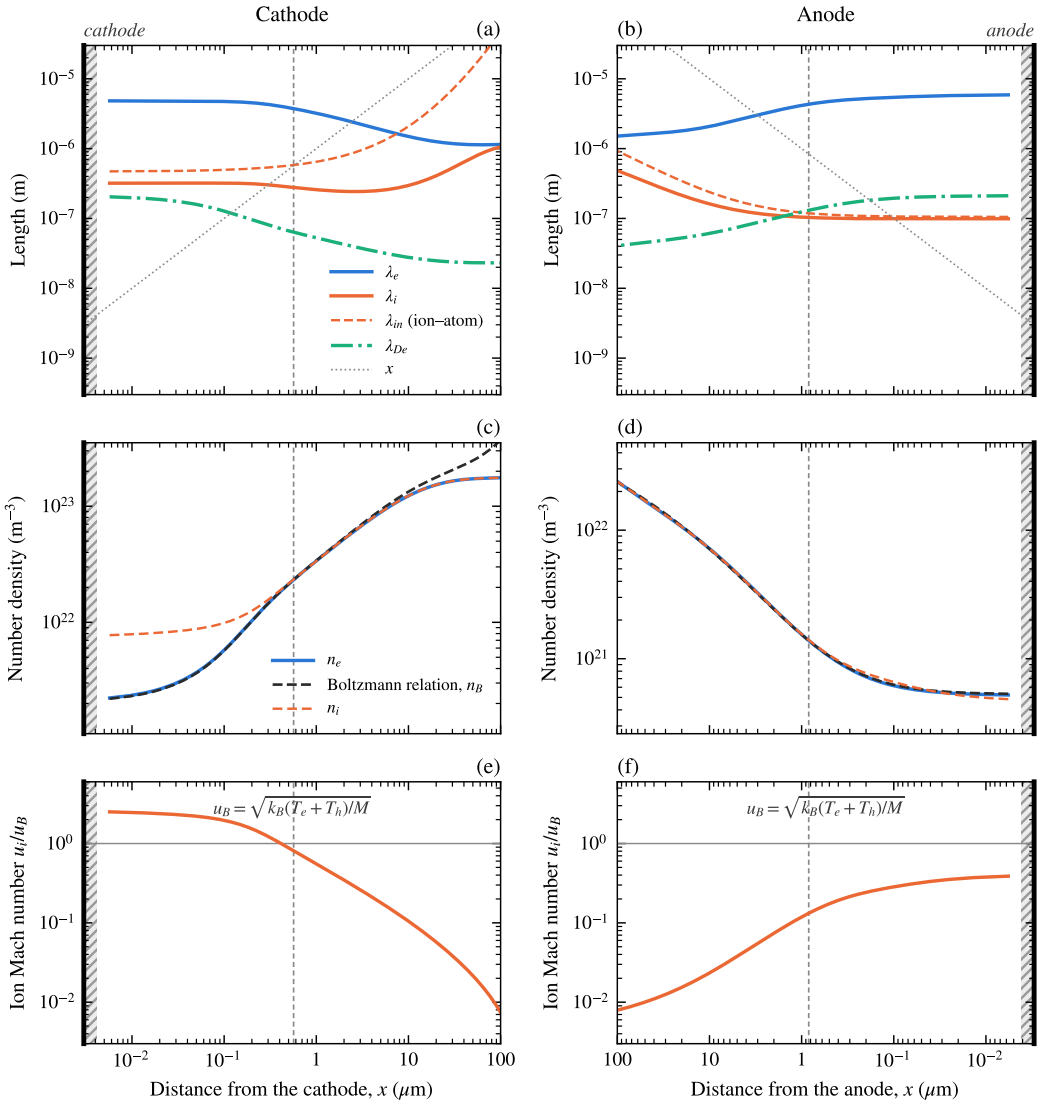



**Figure 6.** The sheaths of the base cases: cathode (left) and anode (right). (a), (b) Electron and ion mean free paths, the ion–atom part of the latter, and the electron Debye length, from the local state; dotted: the distance $x$ from the electrode. (c), (d) Electron density and the Boltzmann relation $n_B = n_e(x_\mathrm{s}) \exp[(e/k_B) \int_{\phi_\mathrm{s}}^{\phi} d\phi'/T_e]$ started at the sheath edge, with the ion density for reference. (e), (f) Ion velocity towards the electrode, $u_i = \Gamma_i/n_i$, relative to the Bohm velocity $u_B = \sqrt{k_B(T_e + T_h)/M}$. The vertical dashed line marks the sheath edge $x_\mathrm{s}$.

and stays above 10 % of the drag out to 2.3 $\mu$m; beyond 14 $\mu$m it is below 1 %. At the anode the inertia is at most 35 % of the drag, at the surface, and below 10 % beyond 0.09 $\mu$m.

*Electrons.* In both sheaths the field retards the electrons, and the electric force on them is balanced by their pressure gradient (figure 5(c), (d)). In the sheath the two cancel to within 0.9 % (cathode) and 4.7 % (anode) of the electric force. The drag (0.6 and 2.6 % of the electric force at the surface) and the thermal force (1.2 and 0.9 %) are small there. The drag becomes important only in the quasi-neutral layer, where it carries the current: at 10 $\mu$m from the surface it is 16 % (cathode) and 20 % (anode) of the electric force. The inertia of the electrons exceeds their drag within 0.33 $\mu$m of the cathode (by up to a factor of 54) and within 0.59 $\mu$m of the anode (by up to a factor of 6), but it remains smaller than the electric force: at the surface it is a third of the electric force at the cathode and 16 % at the anode, and at the sheath edge 0.3 and 2.3 %. At the cathode, this inertia is that of the emitted electrons, accelerated by the sheath field.

*Mean free paths.* Figure 6(a), (b) shows the mean free paths $\lambda_e$ and $\lambda_i$ of table 1, evaluated with the local densities and temperatures, the ion–atom part of $\lambda_i$, $\lambda_{in} = 1/(n_n \overline{Q}_{in})$, and the electron Debye length. The electron mean free path is of order microns, 3.8 $\mu$m at the cathode sheath edge and 4.3 $\mu$m at the anode one. The sheaths, 0.57 and 0.84 $\mu$m thick, therefore contain only 0.13 and 0.17 electron mean free paths, and $\lambda_e$ exceeds the distance from the surface within 2.3 $\mu$m of the cathode and 2.8 $\mu$m of the anode. The ions are more collisional. At the sheath edge, $\lambda_i = 0.28$ $\mu$m at the cathode and 0.10 $\mu$m at the anode, so that the ions cross 1.8 and 8.2 mean free paths in the sheaths (1.05 and 7.4 counting only the collisions with atoms). The anode sheath is thus collision-dominated for the ions, consistent with their small inertia. The cathode sheath is transitional: the ions make one to two collisions across it, and their inertia exceeds the drag over its inner two thirds. There the ions cannot follow the steep rise of the field, and the local relation between field and ion velocity in equation (9) overestimates the ion velocity near the surface. The electrons cross both sheaths practically without collisions. Their balance there is, however, not a balance of collisional transport: the drag is a few per cent of the electric force, and the field and pressure gradient balance each other as in the Boltzmann equilibrium of a Maxwellian population retarded by a potential barrier, which does not require collisions. The electron description therefore remains adequate in the sheaths, except within about 0.1 $\mu$m of the cathode, where the inertia of the emitted electrons is no longer small. Beyond 2–3 $\mu$m from either electrode, the inertia of both species is below 10 % of the drag, and the inertialess, collision-dominated description of section 2.1 holds.

*Boltzmann electrons and the Bohm criterion.* Simplified sheath models describe the electrons in the sheath by the Boltzmann relation and impose the Bohm criterion on the ions entering it, $u_i \geq u_B$, with $u_B = \sqrt{k_B(T_e + T_i)/M}$. Figure 6(c)–(f) tests both against the computed sheaths. The electron density follows the Boltzmann relation closely. Started at the sheath edge, the relation reproduces $n_e$ to within 1.5 % across the cathode sheath, where $n_e$ falls by a factor of 10, and to within 2.1 % across the anode sheath, where it falls by a factor of 2.6. Outside the anode sheath it still holds to within 2.3 % at 100 $\mu$m. At the cathode it fails progressively in the ionization layer, where the electron drag carries the current: $n_e$ is 8 % below $n_B$ at 10 $\mu$m and 53 % below at 100 $\mu$m. The ions, in contrast, do not satisfy the Bohm criterion at the sheath edge (figure 6(e), (f)). At the cathode they enter the sheath at $u_i = 1830$ m s$^{-1}$, against $u_B = 2230$ m s$^{-1}$, i.e. at $0.82\,u_B$. They reach $u_B$ only 0.40 $\mu$m from the surface, inside the sheath, where the charge separation is already 2.9 %, and arrive at the surface at $2.5\,u_B$. At the anode the ions enter the sheath at $0.13\,u_B$ and reach the surface at $0.39\,u_B$, never attaining $u_B$. The Bohm flux $n_i u_B$ at the sheath edge would therefore overestimate the ion flux entering the sheath by 22 % at the cathode and by a factor of 7.5 at the anode. The Bohm criterion is derived for a collisionless sheath. In these sheaths the ions collide with the atoms throughout the presheath and the sheath, and, except in the inner part of the cathode sheath, their velocity is set by the balance of the electric force and the drag rather than by the potential drop they have fallen through. This agrees with the conclusion of Semenov et al. [26] for the anode layer.

### *4.4 Heat flux to the electrode*

The terms of the electrode heat flux (24) are shown in figure 7. At the anode, $q_w = 38.2$ MW m$^{-2}$ and the absorbed electrons deliver 82 % of it: 23.8 MW m$^{-2}$ as the work function released on entering the metal and 7.6 MW m$^{-2}$ as kinetic energy. Gas conduction contributes 6.4 MW m$^{-2}$ and ion recombination about 1 %. The anode heat flux is thus set by the electron current, through $W$, and by the electron temperature at the surface; with the surface temperature prescribed, it is the heat that the anode cooling has to remove.

At the cathode, the heat flux is the small difference of large terms. The absorbed plasma

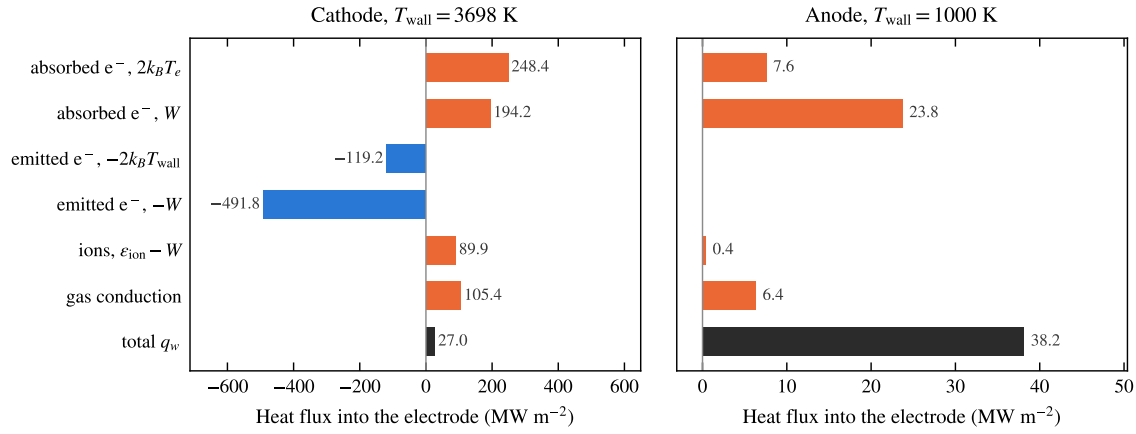



**Figure 7.** Terms of the heat flux delivered to the electrode, equation (24), positive into the electrode, for the cathode (left) and the anode (right). Orange: heating; blue: emission cooling; black: the net heat flux $q_w$. At the cathode, $q_w$ equals the conduction $k_w(T_{\mathrm{wall}} - T_0)/L_w$ into the electrode body, equation (23); the anode surface temperature is prescribed.

electrons (443 MW m$^{-2}$), gas conduction (105 MW m$^{-2}$) and ion recombination (90 MW m$^{-2}$) heat the surface by 638 MW m$^{-2}$ in total, and emission cools it by 611 MW m$^{-2}$, so that only $q_w = 27.0$ MW m$^{-2}$ is conducted into the cathode body. The surface temperature, 3698 K, is therefore fixed mainly by the condition that emission cooling nearly balances the heating, and only weakly by the conduction into the cathode body. Emission cooling grows with $T_{\mathrm{wall}}$ as $T_{\mathrm{wall}}^2 \exp[-(W - \Delta W)/(k_B T_{\mathrm{wall}})]$, i.e. by about 0.27 % per kelvin at this temperature, so a 5 % change in the heating would shift $T_{\mathrm{wall}}$ by only about 20 K, and $q_w$ by less than 1 %. Ion recombination, at $\varepsilon_{\mathrm{ion}} - W = 13.1$ eV per ion, provides 14 % of the heating, although the ions carry only 5.7 % of the current.

*Ion impact energy.* Equation (24) charges the ions with their recombination energy only, not with the kinetic energy with which they strike the surface. In the model, the ion energy follows from the local field through the two-temperature closure (Appendix A), which presumes that the ions make many collisions over the distance in which the field changes. The cathode sheath does not satisfy this: the ions cross it with only one to two collisions, and their inertia exceeds the drag over its inner part (Section 4.3). The local mean ion energy at the cathode surface is therefore not a reliable impact energy, and its distribution would require a kinetic treatment of the sheath. We therefore give only the maximum impact energy at the cathode: that of an ion that enters the sheath with the mean energy at the sheath edge, $\frac{3}{2}k_BT_{\mathrm{eff}} + \frac{1}{2}Mv_d^2 = 1.5$ eV (equations (39) and (40)), and crosses it without a collision, gaining $e\phi_{\mathrm{s}} = 4.1$ eV. This maximum is 5.6 eV per ion, and with the ion flux of table 2 it delivers at most 38 MW m$^{-2}$, comparable to $q_w$ itself. Added to the heating, this term would, however, raise $T_{\mathrm{wall}}$ by only about 20 K, most of it being removed by emission cooling, and $q_w$ by about 1 %. At the anode, where the ions carry 0.4 % of the current and fall through only 0.41 V in a collision-dominated sheath, their kinetic energy is negligible.

### *4.5 Dependence on the bulk temperature and the current density*

The base cases were repeated with the bulk temperature varied at fixed current density, $T_{\mathrm{bulk}} = 11\,000$–$14\,000$ K for the anode and 19 000–23 000 K for the cathode (figure 8), and with the current density varied at the base bulk temperature, $|J_{\mathrm{app}}| = 7 \times 10^6$–$2 \times 10^7$ A m$^{-2}$ for the anode and $8 \times 10^7$–$1.5 \times 10^8$ A m$^{-2}$ for the cathode (figure 9). Everything else is as in the base cases. In particular, $\Lambda$ is recalibrated for each bulk state through equation (14), which gives column radii of 1.0–4.1 mm for the anode cases and 0.33–0.66 mm for the cathode cases.

*Cathode.* The heat flux to the cathode hardly changes: $q_w = 26.3$–$27.3$ MW m$^{-2}$ in every case (figures 8(a) and 9(a)), while $T_{\mathrm{wall}}$ varies only between 3633 and 3727 K. This follows from the heat balance (23). With the electrode parameters used here, a change of $q_w$ by 1 MW m$^{-2}$ requires a change of $T_{\mathrm{wall}}$ by 100 K, which would change the emission cooling by about 30 %. The heating of the surface, by the absorbed electrons, the ions and gas conduction, does depend on the plasma: it increases from 488 to 735 MW m$^{-2}$ between $J_{\mathrm{app}} = 8 \times 10^7$ and $1.5 \times 10^8$ A m$^{-2}$. The surface temperature then rises by only 94 K, and the additional heating is removed by emission cooling. Within these ranges, the heat flux conducted into the cathode is therefore set by the electrode, through $W$, $A_R$, $k_w$, $L_w$ and $T_0$, rather than by the plasma. The plasma side adjusts instead. With

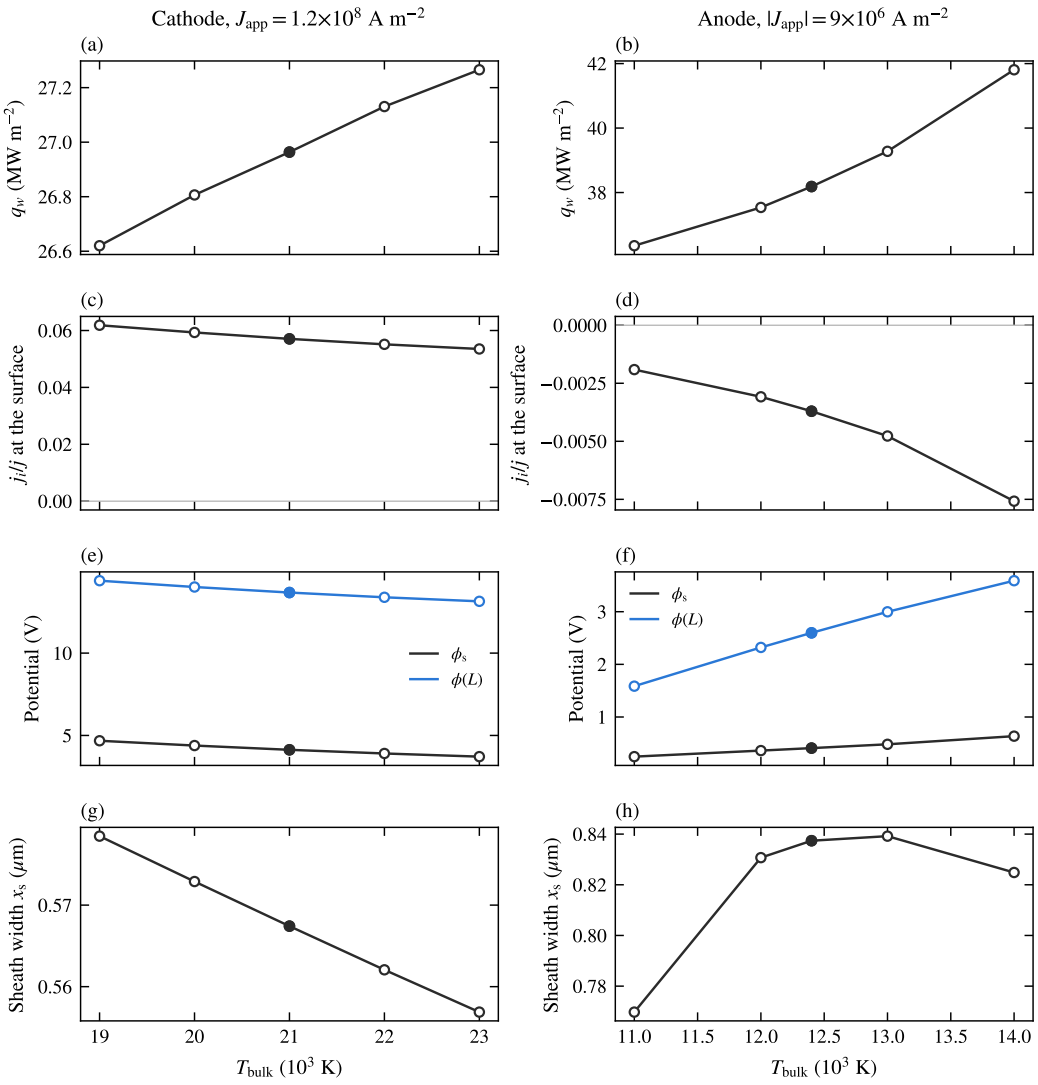



**Figure 8.** Electrode quantities against the bulk temperature $T_{\rm bulk}$, for the cathode at $J_{\rm app} = 1.2 \times 10^8$ A m$^{-2}$ (left) and the anode at $|J_{\rm app}| = 9 \times 10^6$ A m$^{-2}$ (right). Filled symbols: the base cases. (a), (b) Heat flux into the electrode $q_w$, equation (24). (c), (d) Ion fraction of the current density at the surface, positive for ions flowing into the cathode; at the anode the ions also flow into the surface, against the current. (e), (f) Potential at the sheath edge $\phi_{\rm s}$ and at the bulk-plasma boundary $\phi(L)$, relative to the electrode. (g), (h) Sheath width $x_{\rm s}$, the distance from the surface at which $|n_i - n_e|/n_i$ falls to 1 %.

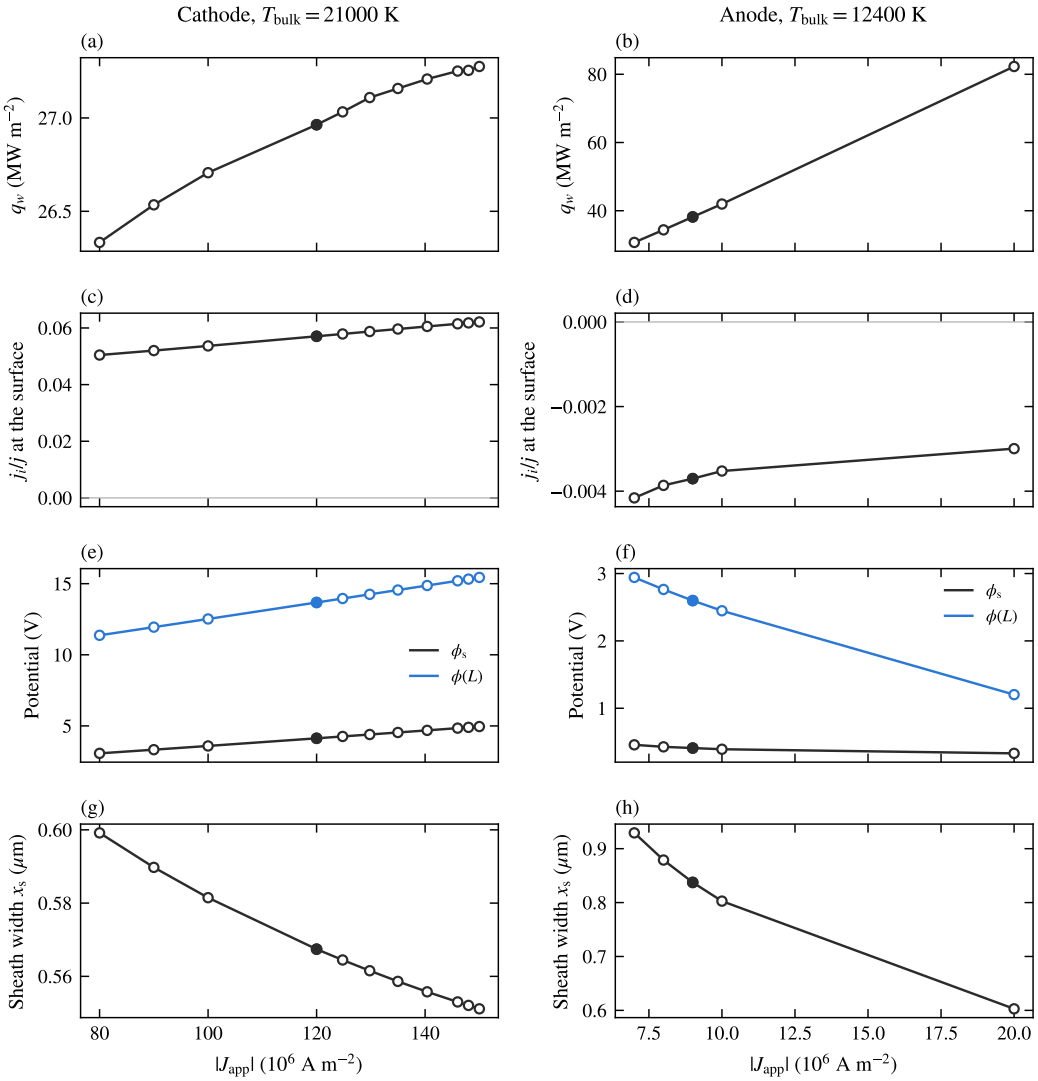



**Figure 9.** As figure 8, against the current density $|J_{\rm app}|$, for the cathode at $T_{\rm bulk} = 21\,000$ K (left) and the anode at $T_{\rm bulk} = 12\,400$ K (right).

increasing current density, the sheath drop rises from 3.1 to 5.0 V and the ion fraction of the current at the surface from 5.0 to 6.2 % (figure 9). The electron temperature at the surface rises from 16.7 to 21.4 kK, and the voltage across the domain from 11.4 to 15.4 V. The emitted current exceeds the arc current in every case, but by less as the current grows, from 1.77 to 1.44 times the arc current. Above $1.5 \times 10^8$ A m$^{-2}$ no steady state was reached with this cathode heat balance. Increasing the bulk temperature at fixed current acts in the opposite direction on the sheath (figure 8). Between 19 000 and 23 000 K, $\phi_s$ decreases from 4.7 to 3.7 V, $j_i/j$ from 6.2 to 5.3 % and $T_e$ at the surface from 20.3 to 18.9 kK, while the voltage across the domain $\phi(L)$ falls from 14.4 to 13.1 V. The sheath width changes by less than 10 % in either sweep (figures 8(g) and 9(g)): it decreases from 0.60 to 0.55 $\mu$m as the current density increases, and from 0.58 to 0.56 $\mu$m as the bulk temperature increases. The larger sheath drop at higher current is held across a nearly unchanged width by a denser ion space charge: the ion density at the surface rises from $5.5 \times 10^{21}$ to $9.2 \times 10^{21}$ m$^{-3}$.

*Anode.* The anode heat flux increases almost in proportion to the current density, from 30.7 to 82.3 MW m$^{-2}$ between $|J_{\text{app}}| = 7 \times 10^6$ and $2 \times 10^7$ A m$^{-2}$ (figure 9(b)), i.e. 4.4 to 4.1 W per ampere. Of this, the work function released by the absorbed electrons contributes exactly $W|J_{\text{app}}|/e$, i.e. 2.63 W per ampere. Their kinetic energy, $2k_BT_e$ per electron with $T_e$ at the surface rising from 4.7 to 6.0 kK, contributes another 0.8–1.0 W per ampere. Gas conduction (6.1–7.9 MW m$^{-2}$) and ion recombination (below 1 MW m$^{-2}$) contribute the rest. At fixed current, the bulk temperature has a much weaker effect, $q_w = 36.4$ to 41.8 MW m$^{-2}$ between 11 000 and 14 000 K (figure 8(b)). The electron terms are fixed by the current and by $T_e$ at the surface, which stays between 4.9 and 5.2 kK, and the increase comes mostly from gas conduction, which rises from 4.4 to 9.1 MW m$^{-2}$. In every anode case, the ions carry less than 1 % of the current. The anode lies below the bulk-plasma boundary by $\phi(L) = 1.6$ to 3.6 V as $T_{\text{bulk}}$ increases from 11 000 to 14 000 K. With increasing current density this drop decreases, from 2.9 V at $7 \times 10^6$ to 1.2 V at $2 \times 10^7$ A m$^{-2}$. The potential at the edge of the thin space-charge layer stays between 0.25 and 0.64 V. The width of this layer follows the electron density at the surface. It decreases from 0.93 to 0.60 $\mu$m as the current density increases from $7 \times 10^6$ to $2 \times 10^7$ A m$^{-2}$ (figure 9(h)), while $n_e$ at the surface rises from $4.1 \times 10^{20}$ to $1.0 \times 10^{21}$ m$^{-3}$, nearly as $n_e^{-1/2}$, the scaling of the Debye length. At fixed current, $n_e$ at the surface changes by at most 3 % with the bulk temperature, and the width stays between 0.77 and 0.84 $\mu$m (figure 8(h)).

### *4.6 Relaxation times*

To estimate how quickly the near-electrode layers follow a change of the arc, each steady state of Sections 4.1–4.5 was perturbed by a step of +2 % in $|J_{\text{app}}|$, and the time-dependent equations were integrated to the new steady state. The surface temperature and the radial-loss coefficient $\Lambda$ were held at their steady values, since the electrode body responds on a time scale of seconds and $\Lambda$ represents the column outside the domain. The response of the ion current density $j_i$ and of the heat flux $q_w$ at the surface is characterized by the time to steady state $t_{99}$, after which the quantity stays within 1 % of its total change.

In a one-dimensional domain, the step also excites an axial acoustic oscillation of the heavy gas between the electrode and the boundary $x = L$. In a three-dimensional arc, such a pressure disturbance is relieved radially within $R/c \approx 0.1$ $\mu$s. The axial acoustics are therefore removed by multiplying the inertia of the heavy gas, the left-hand side of equation (17), by $\epsilon = 10^{-4}$. This raises the sound speed to $c/\sqrt{\epsilon}$ and leaves the steady state, with $\mathbf{u} = 0$, unchanged; $t_{99}$ varies by less than 3 % for $10^{-6} \le \epsilon \le 10^{-3}$. At the anode, the acoustics do not affect the response, and the physical sound speed is retained.

*Cathode.* The response has two stages (figure 10(a)). About 15 % of the change of $j_i$ occurs within 10 ps; the rest follows within about 1 $\mu$s, $t_{99} = 0.98$ $\mu$s, the final approach being exponential with a relaxation time of 0.29 $\mu$s. The heat flux first moves in the opposite direction and settles later, $t_{99} = 1.29$ $\mu$s. The cathode layer responds faster at higher current density: between $8 \times 10^7$ and $1.5 \times 10^8$ A m$^{-2}$, $t_{99}$ falls from 2.1 to 0.62 $\mu$s for $j_i$ and from 2.6 to 0.89 $\mu$s for $q_w$ (figure 10(c)). The bulk temperature changes $t_{99}$ by less than 15 % between 19 000 and 23 000 K (figure 10(e)).

*Anode.* The anode layer relaxes about 40 times more slowly, with a relaxation time of 11 $\mu$s, and its two quantities behave differently (figure 10(b)). Because the dominant electron terms of the heat flux (24) are carried by the current (Section 4.4), 87 % of the change of $q_w$ occurs within 1 ns, and $t_{99} = 20$ $\mu$s. The ion current density first moves in the opposite direction and needs $t_{99} = 48$ $\mu$s. For $j_i$, $t_{99}$ decreases from 74 to 42 $\mu$s as $T_{\text{bulk}}$ rises from 11 000 to 14 000 K and increases only from 47 to 55 $\mu$s over the current range; that of $q_w$ stays between 17 and 20 $\mu$s in all cases (figure 10(d), (f)).

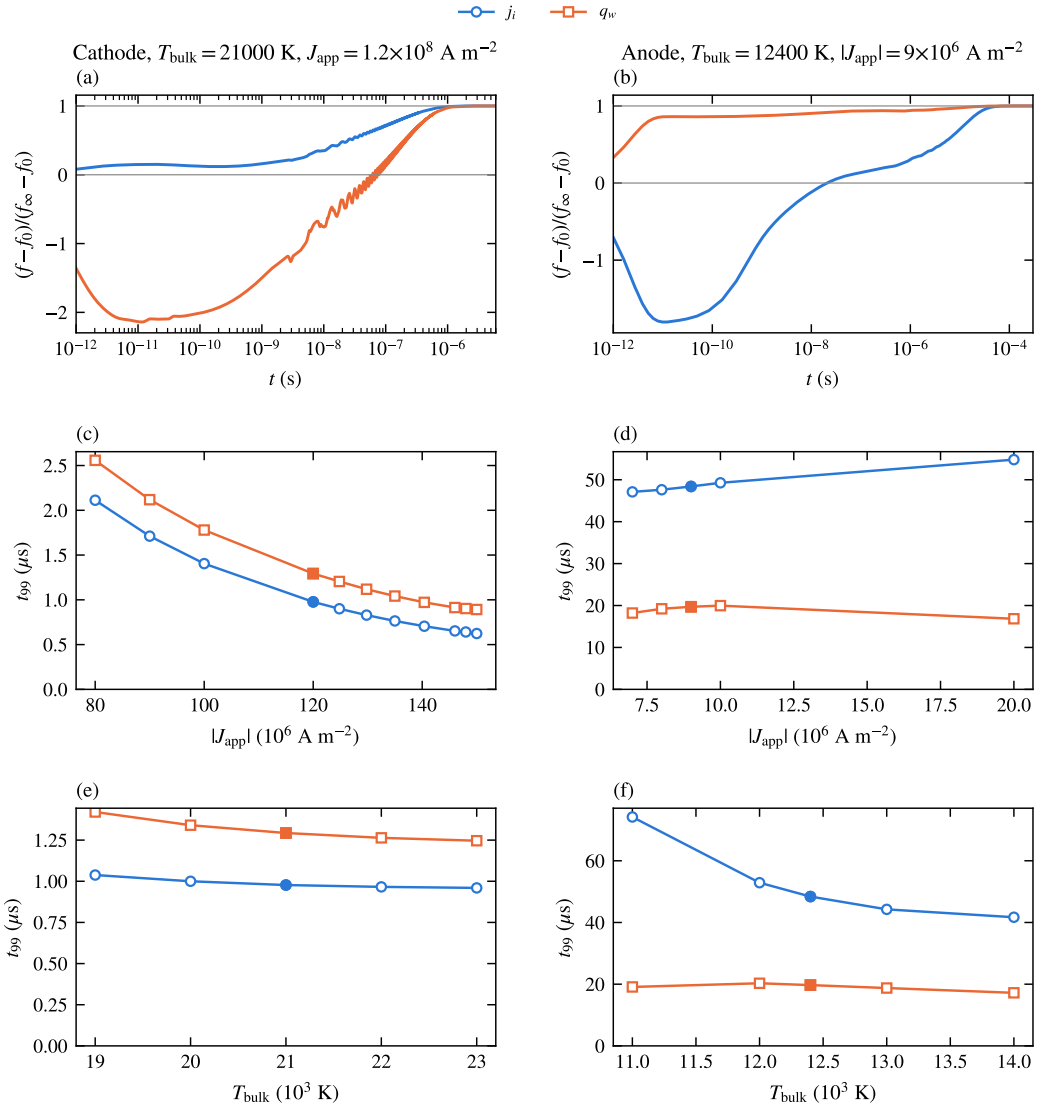



**Figure 10.** Response of the electrode quantities to a +2 % step of $|J_{app}|$: cathode (left, axial acoustics removed) and anode (right). Blue: ion current density $j_i$ at the surface; orange: heat flux $q_w$ into the surface. (a), (b) Normalized response of the base cases, $f_0$ and $f_\infty$ being the initial and the final values. (c), (d) Time to steady state $t_{99}$ against $|J_{app}|$ at the base $T_{bulk}$. (e), (f) $t_{99}$ against $T_{bulk}$ at the base $|J_{app}|$. Filled symbols: the base cases.

The cathode layer can therefore be treated as quasi-steady for changes of the arc slower than a few microseconds. At the anode, the heat flux follows the current within about 20 $\mu$s, but the ion current needs 40–75 $\mu$s.

## 5 Conclusion

A unified drift-diffusion–Poisson model of the near-electrode layers of an atmospheric-pressure argon arc has been presented. A single set of equations is solved from the electrode surface, through the space-charge sheath and the ionization layer, to the bulk plasma: the continuity equations with fluxes derived from the species momentum balances, including thermal diffusion and the collisional drag between species, separate electron and heavy-particle energy equations, a heavy-gas momentum equation and Poisson's equation. Adaptive mesh refinement resolves the sheath on a fraction of the Debye length. The surface temperature of the cathode follows from a heat balance that includes thermionic emission with Schottky lowering. The model reproduces the near-cathode layer computed by Almeida et al. [24] and the near-anode layer computed by Semenov et al. [26]: the temperatures agree within 3–5 %, the charged-particle densities within 15 % outside the cathode sheath, and the ion fraction of the current at the surface within 5 %.

For a cathode at $T_{\mathrm{bulk}} = 21\,000$ K and $1.2 \times 10^8$ A m$^{-2}$, and an anode at 12 400 K and $9 \times 10^6$ A m$^{-2}$, with the radial heat loss of the column calibrated on the bulk state, the main results are as follows.

- The two layers have opposite structures. The cathode has an ion-rich sheath 0.57 $\mu$m wide with a drop of 4.1 V, followed by an ionization layer of about $10^{-4}$ m in which the gas is heated and the plasma becomes almost fully ionized. Joule heating is the dominant electron energy source throughout. The anode has practically no sheath (0.41 V across 0.84 $\mu$m). Its 2.6 V retarding drop is spread over the quasi-neutral layer, the charged particles are supplied by diffusion at densities far above ionization equilibrium, and $T_e - T_h$ reaches 3.9 kK at the surface. The electrons lose energy to the retarding field, and this loss is supplied by the electron energy flux from the column, mainly through conduction and the thermal-diffusion heat flux close to the anode.

- The electrons cross both sheaths with fewer than 0.2 collisions, yet their density follows the Boltzmann relation within about 2 %, so that the electron force balance remains adequate there. The ions cross 1.8 (cathode) and 8.2 (anode) mean free paths in the sheaths and do not satisfy the Bohm criterion at the sheath edge, where they move at 0.82 and 0.13 times the Bohm velocity. A Bohm flux at the sheath edge would overestimate the ion flux by 22 % at the cathode and by a factor of 7.5 at the anode.

- The cathode heat flux, 27 MW m$^{-2}$, is the small difference between the heating of the surface (638 MW m$^{-2}$) and emission cooling (611 MW m$^{-2}$). Because emission cooling depends steeply on $T_{\mathrm{wall}}$, the heat flux into the cathode body is set by the electrode rather than by the plasma: it changed by less than 4 % over all the cathode cases, while the heating of the surface changed by 50 %. The anode heat flux, 38 MW m$^{-2}$, is 82 % electronic and increases nearly in proportion to the current, at 4.1–4.4 W per ampere, of which 2.63 W per ampere is the work function.

- Because the ions cross the cathode sheath with only one to two collisions, a local estimate of their impact energy is not reliable, and only its maximum, for an ion crossing the sheath without a collision, has been given: 5.6 eV, corresponding to at most 38 MW m$^{-2}$. Owing to the stiffness of emission cooling, this term would change the cathode surface temperature by only about 20 K.

- Over the ranges of bulk temperature and current density studied, the cathode sheath width stays between 0.55 and 0.60 $\mu$m, its larger drop at higher current being held by a denser ion space charge. The anode sheath width follows the Debye length at the surface and decreases from 0.93 to 0.60 $\mu$m as the current density increases from $7 \times 10^6$ to $2 \times 10^7$ A m$^{-2}$.

- After a change of the current, the cathode layer reaches a new steady state within 0.6–2.6 $\mu$s. At the anode, the heat flux follows within about 20 $\mu$s, but the ion current needs 40–75 $\mu$s.

The local, inertialess transport description is marginal within about 0.4 $\mu$m of the cathode, where the inertia of the ions exceeds their drag, and the ion energy distribution at the cathode requires a kinetic treatment of the sheath. The present one-dimensional calculations represent the radial losses of the column through a calibrated sink. Since the layers settle within microseconds and are characterized by the bulk state and the current density, the model can supply electrode boundary conditions, such as the voltage drop, the current partition and the heat flux to the electrode, to LTE models of the arc column.

## A Transport coefficients

This appendix describes how the transport and energy-exchange coefficients that close the model are calculated.

### *A.1 Electrons*

Following Mitchner and Kruger [39], the electron current density $\mathbf{J}_e = -e\boldsymbol{\Gamma}_e$ and the electron heat flux $\mathbf{q}_e$ are written as

$$\mathbf{J}_e = \sigma\left(\mathbf{E} + \nabla p_e/(en_e)\right) + \varphi\,\nabla T_e, \tag{27}$$

$$\mathbf{q}_e = -\lambda \nabla T_e + \psi\,\mathbf{J}_e. \tag{28}$$

Here $\sigma$ is the electrical conductivity, $\varphi$ is the thermal-diffusion coefficient, $\lambda$ is the ordinary (zero-current) thermal conductivity, and $\psi$ is the energy transported by the current per unit charge. Comparing equations (27) and (28) with the electron momentum balance (2) and the energy flux (12) gives

$$\bar{\nu}_e^M = \frac{n_e e^2}{m_e \sigma}, \qquad C_e = \frac{e}{k_B}\frac{\varphi}{\sigma}, \qquad \kappa_e = \lambda, \qquad -\frac{e\psi}{k_B T_e} = \frac{5}{2} + C_e. \tag{29}$$

Mitchner and Kruger [39] give $\sigma$, $\varphi$, and $\lambda$ in the weakly and fully ionized limits. Substituting them into equation (29) gives the collision frequencies and thermal-diffusion coefficients in each limit.

*Weakly ionized limit.* Electron–neutral collisions dominate, and the electrons behave as a Lorentz gas whose distribution function has a Maxwellian isotropic part $f_M(T_e)$. Let $C$ be the electron speed, $\varepsilon = \frac{1}{2}m_e C^2$ the electron energy, $x = \varepsilon/(k_B T_e)$ the reduced energy, and $\nu_{en} = n_n C Q^{(1)}(\varepsilon)$ the energy-dependent momentum-transfer frequency, where $Q^{(1)}$ is the momentum-transfer cross section. A two-term solution of the electron Boltzmann equation then gives

$$\bar{\nu}_{en}^M = \frac{\langle C^2\rangle_0}{\langle C^2/\nu_{en}\rangle_0}, \qquad C_{en} = \langle x\rangle_\nu - \frac{5}{2}, \qquad \lambda_{en} = \frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{en}^M}\left[\langle x^2\rangle_\nu - \langle x\rangle_\nu^2\right], \tag{30}$$

where $\bar{\nu}_{en}^M$ is the momentum-transfer average of $\nu_{en}$, $\langle\cdot\rangle_0$ is the Maxwellian average, and $\langle\cdot\rangle_\nu$ is the transport average with the weight $C^4 f_M/\nu_{en}$:

$$\langle g\rangle_0 = 4\pi\int_0^\infty g\, C^2 f_M\, dC, \qquad \langle g\rangle_\nu = \frac{\langle g\, C^2/\nu_{en}\rangle_0}{\langle C^2/\nu_{en}\rangle_0}. \tag{31}$$

The elastic energy exchange in equations (11) and (15) uses instead the energy-exchange average

$$\bar{\nu}_{en}^E = \frac{\langle C^2 \nu_{en}\rangle_0}{\langle C^2\rangle_0}. \tag{32}$$

All averages are computed from the argon momentum-transfer cross section $Q^{(1)}$ of the LXCat database [40].

*Fully ionized limit.* For singly charged ions, the Spitzer–Härm solution gives [39]

$$\bar{\nu}_{ei}^M = \frac{\bar{\nu}_{ei}^E}{1.975}, \qquad C_{ei} = \frac{1.389}{1.975} = 0.703, \qquad \lambda_{ei} = 3.203\,\frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{ei}^E}, \tag{33}$$

where $\bar{\nu}_{ei}^E$ is the energy-exchange average (32) of the Coulomb collision frequency [39],

$$\bar{\nu}_{ei}^E = n_i\,\frac{4\sqrt{2\pi}}{3}\left(\frac{m_e}{k_B T_e}\right)^{3/2}\left(\frac{e^2}{4\pi\varepsilon_0 m_e}\right)^2 \ln\Lambda, \tag{34}$$

and $\ln\Lambda$ is the Coulomb logarithm,

$$\ln\Lambda = \max\left(2,\ \ln\frac{\lambda_{De}}{b_{90}}\right), \qquad \lambda_{De} = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_e e^2}}, \qquad b_{90} = \frac{e^2}{12\pi\varepsilon_0 k_B T_e}. \tag{35}$$

Here $\lambda_{De}$ is the electron Debye length and $b_{90}$ is the 90°-deflection impact parameter. The floor at 2 keeps $\ln\Lambda$, and hence $\bar{\nu}_{ei}^E$, positive when the plasma is not weakly coupled.

At intermediate ionization, the two limits are combined. The momentum-transfer frequencies are added, the thermal conductivities are combined in series (their thermal resistances add), and the thermal-diffusion coefficient is the mean of the two limits weighted by the energy-exchange frequencies:

$$\bar{\nu}_e^M = \bar{\nu}_{en}^M + \bar{\nu}_{ei}^M, \qquad \frac{1}{\kappa_e} = \frac{1}{\lambda_{en}} + \frac{1}{\lambda_{ei}}, \qquad C_e = \frac{C_{en}\bar{\nu}_{en}^E + C_{ei}\bar{\nu}_{ei}^E}{\bar{\nu}_{en}^E + \bar{\nu}_{ei}^E}. \tag{36}$$

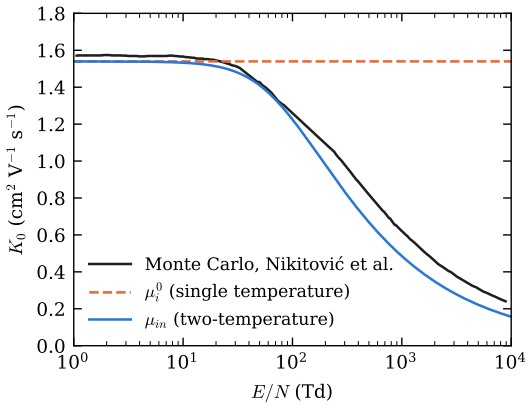



**Figure 11.** Reduced mobility of $Ar^+$ in Ar at 300 K as a function of the reduced field $E/N$. The single-temperature mobility $\mu_i^0$, equation (38) (dashed), and the two-temperature mobility $\mu_{in}$, equation (40) (solid blue), are compared with the Monte Carlo results of Nikitović and Raspopović [44] (black), digitized from their figure 7.

### *A.2 Ions*

The ion–neutral transport is governed by resonant charge exchange. The momentum-transfer cross section is taken from the fit [32]

$$\overline{Q}_{in} = 2\left[7\times10^{-10} - 0.6\times10^{-10}\ln\left(\frac{2k_BT_h}{e}\right)\right]^2 \ \mathrm{m}^2, \tag{37}$$

with $2k_BT_h/e$ in eV. The corresponding low-field mobility is obtained from the Mason–Schamp equation [41],

$$\mu_i^0 = \frac{1.104\times10^5}{\overline{Q}_{in}\sqrt{T_h}\,n_n} \ \mathrm{m^2\,V^{-1}\,s^{-1}} \qquad (\overline{Q}_{in} \text{ in m}^2,\ T_h \text{ in K},\ n_n \text{ in m}^{-3}). \tag{38}$$

To account for the effect of strong electric fields on the ion mobility, a closed-form two-temperature Wannier formulation [42, 43] is used, in which the drift speed $v_d$ and the effective ion temperature $T_{\mathrm{eff}}$ are determined self-consistently:

$$v_d^2 = \frac{\sqrt{1+4\mathcal{A}\mathcal{B}}-1}{2\mathcal{B}}, \qquad \mathcal{A} = \left(\mu_i^0|\mathbf{E}|\right)^2, \qquad \mathcal{B} = \frac{M}{3k_BT_h}, \tag{39}$$

$$T_{\mathrm{eff}} = T_h + \frac{Mv_d^2}{3k_B}, \qquad \mu_{in} = \mu_i^0\sqrt{\frac{T_h}{T_{\mathrm{eff}}}}. \tag{40}$$

The ion–neutral momentum-transfer frequency is then obtained from the mobility as

$$\bar{\nu}_{in}^M = \frac{e}{M\mu_{in}}. \tag{41}$$

Figure 11 compares the reduced mobility $K_0 = \mu n_n/N_0$ (with $N_0 = 2.687\times10^{25}$ m$^{-3}$) of the single-temperature mobility $\mu_i^0$, equation (38), and the two-temperature mobility $\mu_{in}$, equation (40), with the Monte Carlo results of Nikitović and Raspopović [44] for $Ar^+$ in Ar at 300 K. These Monte Carlo results agree with the measured drift-tube mobilities [44]. At $T_h$ fixed, $K_0$ depends only on $E/N$. At low field, both formulations give $K_0 = 1.54$ cm$^2$ V$^{-1}$ s$^{-1}$, 2 % below the Monte Carlo value. Above about 30 Td, the single-temperature mobility stays constant and overestimates $K_0$ by a factor of 1.6 at 300 Td and 4 at 3000 Td. The two-temperature mobility reproduces the decrease of $K_0$ with $E/N$, and lies 15 % below the Monte Carlo value at 300 Td and 26 % below at 3000 Td. Most of this remaining difference comes from evaluating $\overline{Q}_{in}$ at $T_h$ instead of $T_{\mathrm{eff}}$. Evaluating it at $T_{\mathrm{eff}}$ brings the two-temperature mobility within 7 % of the Monte Carlo values from 1 to 7000 Td, but then $T_{\mathrm{eff}}$ has to be found iteratively instead of in closed form.

The ion–electron frequency follows from momentum conservation, equation (5),

$$\bar{\nu}_{ie}^M = \frac{m_e n_e}{M n_i}\,\bar{\nu}_{ei}^M, \tag{42}$$

with $\bar{\nu}_{ei}^{M}$ given by equation (33). The ions take up the electron–ion part of the electron thermal force, so their thermal coefficient is the $C_{ei}$ term of equation (36),

$$C_i = \frac{C_{ei}\bar{\nu}_{ei}^{E}}{\bar{\nu}_{en}^{E} + \bar{\nu}_{ei}^{E}}. \tag{43}$$

The ion contribution $\kappa_i$ to the heavy-particle thermal conductivity is calculated as in Almeida et al. [24].

### *A.3 Heavy species*

The atom frequencies follow from momentum conservation, equation (5),

$$\bar{\nu}_{ni}^{M} = \frac{n_i}{n_n}\,\bar{\nu}_{in}^{M}, \qquad \bar{\nu}_{ne}^{M} = \frac{m_e n_e}{M n_n}\,\bar{\nu}_{en}^{M}. \tag{44}$$

The atoms take up the electron–neutral part of the electron thermal force, so their thermal coefficient is the $C_{en}$ term of equation (36),

$$C_n = \frac{C_{en}\bar{\nu}_{en}^{E}}{\bar{\nu}_{en}^{E} + \bar{\nu}_{ei}^{E}}, \tag{45}$$

and equations (43) and (45) satisfy $C_i + C_n = C_e$, as required by equation (5). The heavy-particle thermal conductivity $\kappa_h = \kappa_n + \kappa_i$, the sum of the atom and ion contributions, is calculated as in Almeida et al. [24].

The electrical conductivity $\sigma = n_e e^2/(m_e \bar{\nu}_e^{M})$ and the thermal conductivity obtained with these coefficients are compared with the transport properties of argon in local thermodynamic equilibrium (LTE) in figure 12. The coefficients are evaluated with $T_e = T_h = T$ and the Saha composition at atmospheric pressure, and compared with the data tabulated by Boulos et al. [45]. Along the axis, the model has no reactive thermal conductivity: the energy carried by the diffusion of ion pairs and released on recombination appears through the ambipolar flux and the ionization source ($\kappa_R$ enters only the radial loss (13)). For the comparison, the reactive conductivity $\kappa_R$ computed with Mutation++ [37] is therefore added to $\kappa_e + \kappa_h$.

The electrical conductivity agrees well with the LTE data over most of the temperature range, with only minor deviations in the low-temperature regime where electron–neutral collisions dominate. At intermediate temperatures, however, the additive rule for the collision frequencies overestimates the conductivity. This discrepancy arises because, in argon, the slow electrons associated with the Ramsauer minimum contribute substantially to the current in the electron–neutral limit, while these same electrons are strongly scattered by ions. Simply adding the averaged electron–neutral and electron–ion collision frequencies therefore fails to capture the correlation between the two collision processes. The thermal conductivity also shows good agreement with the reference data in the high-temperature regime, with a modest overprediction at intermediate temperatures.

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To be completed.

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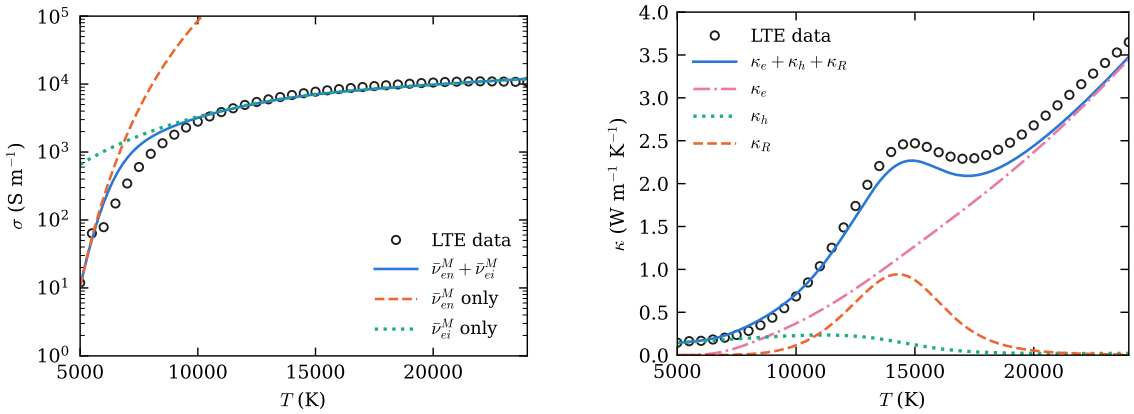



**Figure 12.** Transport coefficients of argon in LTE at atmospheric pressure, computed with the rules of equation (36) (lines), compared with the LTE data of Boulos et al. [45] (symbols). Left: electrical conductivity $\sigma = n_e e^2/(m_e \bar{\nu}_e^M)$, with the electron–neutral and electron–ion limits shown separately. Right: thermal conductivity, the sum of the electron conductivity $\kappa_e$, the heavy-particle conductivity $\kappa_h$ [24], and the reactive conductivity $\kappa_R$ computed with Mutation++ [37].


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